A13 GST TRANSMITTER

ALIGN

INSTRUCTION MANUAL

遙控器使用說明書

[HERA1301T]



14 CHANNELS

2.4GHz TRANSMITTER

For more detailed instruction, please check our website.
http://www.align.com.tw/A13/
更多詳細說明請上官網。

Congratulations on your purchase of Align professional aerial photography products! To ensure your success with this product, we would like to present the following information and important reminders.

承蒙閣下選用亞拓遙控世界系列產品，謹表謝意。進入遙控世界之前必須告訴您許多相關的知識與注意事項，以確保您能夠在使用的過程中較得心應手。在開始操作之前，請務必詳閱本說明書，相信一定能夠給您帶來相當大的幫助，也請您妥善保管這本說明書，以作為日後參考。

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INTRODUCTION
前言

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Thank you for purchasing the A13 GST transmitter digital proportional R/C airplane/helicopter system. This transmitter has been designed and manufactured to provide you with many, many years of modeling enjoyment and fun. Whether this is your first venture into R/C or you're a seasoned veteran, the A13 GST transmitter offers something for everyone. If this is your first "computer" radio, rest assured that this system is designed to make initial setup and field-tuning of your airplane and/or helicopter easier and more accurate than using a non-computer "analog" radio. Experienced modelers will appreciate some of the A13 GST's advanced features and functions normally found on the more advanced radios. In either case, to make the best use of the A13 GST system, and to operate it safely, you should carefully read the instructions in their entirety prior to operation.

Every modeler has their own personal preferences on the proper steps to set-up and program their models. As such, it would be very difficult to create a manual that would satisfy all modelers accordingly. Thus, this manual has been designed to 'step you through' the programming of the A13 GST transmitter, following the function list of the system. If you wish to follow your own personal preference on setting up a model, please feel free to do so keeping in mind that some programming options will interact with others. For example, Dual Rates and End Points will influence one another.

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每個模型使用者都會依自己的個人喜好採適當的步驟及設置其模型。因此，創造一本可以滿足所有模型使用者的手冊是非常困難的一件事情。致此，透過本手冊，能讓您“按步就班“設定A13 GST遙控器的各項功能。如果您想依個人喜好建立一個新模型，請記住一些會相互影響的功能設定選項。例如，大小動和舵機行程量這兩功能是會有連帶關係的。

SUGGESTION
建議

If, while reading the instructions, you are unclear of some of the procedures or functions described or become 'stuck', please continue reading anyway. Oftentimes, the function or procedure will be explained again later in an alternate manner which will clarify the procedure or function. Another alternative is to connect the battery, servos, receiver, etc. to simulate your aircraft and actually program the transmitter accordingly. This will allow you to visually recognize the effects of your programming inputs. If utilizing this alternate method of familiarizing yourself with your transmitter, please ensure that your aircraft can not cause harm to yourself or others by disabling it accordingly.

This product is to be used for the flying of transmitter controlled models only. ALIGN is not responsible for the results of use of this product by the customer or for any alteration of this product including modification or incorporation into other devices by third parties. Modification will void any warranty and is done at the owner's risk and responsibility. Please protect the environment by disposing of rechargeable batteries responsibly. Throwing rechargeable batteries into the trash or municipal waste system is illegal in some areas.

如果，您對此遙控器的程式指令或功能不是很瞭解而發生困惑時，請查閱本說明書。通常，有些功能或程式在另一個環境中才會被使用。當您改變設定時，建議要連接電池、開關和伺服器、接收器等，並且安裝在實際的模型上，這樣有助於展現改變設定後的實際效果。本產品僅限於無線電遙控模型飛行使用。肇因於改裝或使用由第三方製造的產品設備所產生的產品問題，本公司將不負任何責任。擅自拆裝等不當改裝將不受任何保固以及須自負其所衍生的責任。

為保護環境，對充電電池的處理是大家應負的責任。在某些地區的垃圾處理系統上，將充電池扔進垃圾桶是違法的行為，請知悉有關您所在地區的鎳鎘電池回收方式，並依各所在地區相關法令規範辦理。

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IMPORTANT NOTES
重要聲明

Radio Control (R/C) multicopters are not toys. R/C multicopters utilize various high-tech components to achieve superior performance. Improper use of this product can result in serious injury or even death. Please read this manual carefully before operating, and make sure to be conscious of your own personal safety and the safety of others nearby when operating all ALIGN products. Manufacturer and seller assume no liability for the operation or the use of this product. This product is intended for use only by adults with experience flying remote control aircraft at legal flying fields. After the sale of this product we cannot be held liable over its operation or usage.

We recommend that you seek the assistance of an experienced pilot before attempting to fly our products for the first time. A local expert is the best way to properly assemble, setup, and fly your model for the first time. This product requires a certain degree of skill to operate, and is an expendable item. Any damage or dissatisfaction as a result of accidents or modifications are not covered by any warranty and cannot be returned for repair or replacement. Please contact our distributors for free technical consultation and parts at discounted rates when you experience problems during operation or maintenance. As Align Corporation Limited has no control over the use, setup, assembly, modification or misuse, no liability shall be assumed nor accepted for any resulting damage or injury. By the act of use, setup or assembly, the user accepts all resulting liability.

In addition, R/C multicopters and its components are precision electronics susceptible to interferences from external forces such as magnetic field and transmitter signal. Should the multicopter or any onboard photographic equipment suffers loss or crash damage as result of external magnetic or transmitter interferences, Align cannot be held liable as the cause is beyond our control.

As the user of this product, you are solely responsible for operating in a manner that does not endanger yourself and others or result in damage to the property of others.

遙控模型包括遙控器、電子設備、遙控飛行機並非玩具，它是結合了許多高科技產品所設計出來的休閒用品，所以商品的使用不當或不熟悉都可能會造成嚴重傷害甚至死亡，使用之前請務必詳讀說明書，勿輕忽並注意自身安全。注意！任何遙控模型的使用，製造商和經銷商是無法對使用者於零件使用的損耗異常或組裝不當所發生之意外負任何責任，本產品是提供給有操作過遙控飛行機經驗的成人或有相當技術的人員在旁指導，並於當地合法遙控飛行場飛行，以確保安全無虞下操作使用。產品售出後本公司將不負任何操作和使用控制上的任何性能與安全責任。

遙控模型屬於需高操作技術且為消耗性之商品，如經拆裝使用後，會造成不等情況零件損耗，任何使用情況所造成商品不良或不滿意，將無法於保固條件內更換新品或退貨，如遇有使用操作維修問題，本公司全省分公司或代理商將提供技術指導、特價零件供應服務。對使用者的不當使用、設定、組裝、修改、或操作不良所造成的破損或傷害，本公司無法控制及負責。且遙控飛行機與配件之精密電子產品，易受外力、磁場、訊號干擾，在使用過程中如外力、磁場、訊號干擾，導致飛行機本身、及其搭載之攝影設備、器材之損壞或滅失，本公司亦無法控制及負責。

做為本產品的使用者，您，是唯一對於您自己操作的環境及行為負全部的責任之人。

This product complies with low-power transmitter wave radiation of the provisions of motor management :

- Article 12: Without permission granted by the DGT, any company, enterprise, or user is not allowed to change frequency, enhance transmitting power or alter original characteristic as well as performance to approved low power transmitter-frequency devices.
- Article 14: The low power transmitter-frequency devices shall not influence aircraft security and interfere legal communications; If found, the user shall cease operating immediately until no interference is achieved.
- The said legal communications means transmitter communications is operated in compliance with the Telecommunications Act. The low power transmitter-frequency devices must be susceptible with the interference from legal communications or ISM transmitter wave radiated devices.

本產品符合低功率電波輻射性電機管理辦法：

- 第十二條 :經形式認證合格之低功率射頻電機，非經許可，公司、商號或使用者均不得擅自變更頻率、加大功率或變更原設計之特性及功能。
- 第十四條 :低功率射頻電機之使用不得影響飛航安全及干擾合法通信；經發現有干擾現象時，應立即停用，並改善至無干擾時方得繼續使用。
- 前項合法通信，指依電信規定作業之無線電信。低功率射頻電機須忍受合法通信或工業、科學及醫療用電波輻射性電機設備之干擾。

Warning: changes or modifications not expressly approved by the party responsible for compliance could void the user's authority to operate the equipment.

Note: This equipment has been tested and found to comply with the limits for a Class B digital device, pursuant to Part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference in a residential installation. This equipment generates, uses and can radiate transmitter frequency energy and, if not installed and used in accordance with the instructions, may cause harmful interference to transmitter communications. However, there is no guarantee that interference will not occur in a particular installation.

If this equipment does cause harmful interference to transmitter or television reception, which can be determined by turning the equipment off and on, the user is encouraged to try to correct the interference by one or more of the following measures:

- Reorient or relocate the receiving antenna.
- Increase the separation between the equipment and receiver.
- Connect the equipment into an outlet on a circuit different from that to which the receiver is connected.
- Consult the dealer or an experienced transmitter/TV technician for help.

警告：未經允許或依規範改裝或修改此設備規格，可能會影響使用者操作此設備的權利。

注意：本設備已通過測試且符合FCC條款第15條B類電子設備的規範。 此規範主要為保護電子設備防範在安裝使用過程中受到其它訊號干擾。此設備使用時會產生射頻能量，如果未依指示規範安裝或使用，將可能對無線電通信造成有害干擾。但無法保證在安裝時不會發生無線射頻干擾。

假設此設備透過開關切換證實對無線電或電視接收產生有害干擾，建議使用者參考以下修正方式：

- 重新調整或擺放接收天線。
- 增加設備和接收器之間的距離。
- 將設備與連接器連接到不同的電源電路插座。
- 請洽經銷商或經驗充足的技術人員尋求協助。

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SAFETY NOTES
安全注意事項

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WARNING LABEL LEGEND
標誌代表涵義

FORBIDDEN 禁 止	Do not attempt under any circumstances. 在任何禁止的環境下，請勿嘗試操作。
WARNING 警 告	Mishandling due to failure to follow these instructions may result in serious damage or injury. 因為疏忽這些操作說明，而使用錯誤可能造成財產損失或嚴重傷害。
CAUTION 注 意	Mishandling due to failure to follow these instructions may result in danger. 因為疏忽這些操作說明，而使用錯誤可能造成危險。

- Fly only in safe areas, away from other people. Do not operate R/C aircraft indoors or within the vicinity of homes or crowds of people. R/C aircraft are prone to accidents, failures, and crashes due to a variety of reasons including: lack of maintenance, pilot error, and transmitter interference. Pilots are responsible for their actions and damage or injury occurring during the operation or as of a result of R/C aircraft models.
- Prior to every flight, carefully check all parts such as blades, screws, frame, arms, etc; ensure they are firmly secured and show no unusual wears, or unforeseen danger may happen.
- Please obey all regulations to enjoy safe modeling.
- Please keep the model in sight at all times as large objects can negatively impact the RF signal. Please keep in mind that objects such as wire fences and wire mesh will also cause degradation of the RF signal.

●遙控模型屬高危險性商品，操作時務必遠離人群。人為組裝不當或未定期檢修造成的機件損壞、電子控制設備不良，以及操控上的不熟悉、都有可能導致模型失控損傷等不可預期的意外，請操作者務必注意操作安全，並需了解自負疏忽所造成任何意外之責任。

●每趟飛行前須仔細檢查遙控模型機身各部位之零/配件/電子設備之性能是否正常，及無損耗老化現象，並確實將螺絲鎖緊才能升空飛行。並做好定期檢修，避免零件或電子產品異常所造成不可預期意外。

●請遵守所有規定以享受安全的模型活動。

●請在任何時候都將視線保持在模型上，大型物件會對射頻信號產生負面的影響。請牢記，如鐵絲圍欄和鐵絲網等物件也將引起發射頻信號衰減。

FORBIDDEN 禁 止

- Do not use the product at night or in bad weather like rain or thunderstorm. It can cause erratic operation or loss of control.
- Do not use the product in poor visibility conditions.
- Do not use the product on rainy or snowing days. Any exposure to moisture (like water or snow), erratic operation or loss of control may occur.
- Frequency interference may cause loss of control, make sure not to use the products in the following places to ensure the safety of yourself :
  1. Near the base station or other transmitter wave exposure places
  2. Near high voltage cables and broadcast communication antennas.
  3. Near crowd or on the streets/roads where vehicle pass by.
  4. On open water area where passenger ship go through.

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- Do not use this product when you feel tired, unwell, or taking drugs that are alcoholic, or cause anesthesia or excitement, as this may cause serious dangerous to yourself or others.
- The 2.4 GHz transmitter wave is completely different from the low frequency transmitter band as previously. When using, please make sure to keep products within your sight cause large obstacles will block transmitter frequency transmission and result in failure or out of control causing unpredictable danger and loss.
- While using, ensure not to hold transmitter antenna in hands or it will reduce the quality and intensity of transmitter transmission causing model out of control.
- After landing, ensure not to touch any electrons such as motor, ESC or starter to prevent serious scald or burn.

FORBIDDEN 禁 止

- 請不要在夜間或雷雨天氣使用本產品，惡劣的天氣環境有可能導致遙控設備失靈。
- 請不要在能見度有限的情況下使用本產品。
- 請不要雨雪或有水的地方使用本產品。如果有液體進入到系統內部，可能會導致信號干擾可能導致設備失控。
- 為確保您和他人的安全，請不要在以下地點使用本產品
  1. 基地台附近或其他無線電活躍的地方。
  2. 高壓電線或通訊廣播天線附近。
  3. 人群衆多場所或道路有車輛通行的附近。
  4. 有客船運輸通行的水域。
- 當您感到疲倦、身體不適、或服用含有酒精性、或會導致麻醉或興奮的藥物後，嚴禁使用本產品，否則可能對自己或他人造成嚴重的傷害。
- 2.4GHz無線電波完全不同於之前所使用的低頻無線電波段。使用時請確保模型產品在您的視線範圍內，較大的障礙物將會阻斷無線電頻率信號，進而發生遙控失靈及模型失控，導致人員生命傷害及財務的損失。造成不可預期的危險。
- 在使用過程中，嚴禁緊握遙控器天線，否則將會減弱無線電傳播訊號的質量和強度，導致遙控失靈及模型失控。
- 在操作或使用遙控模型後，請勿觸任何可能發熱的部位，例如啓動機、馬達、定速設定等，以避免造成嚴重的燒燙傷。

WARNING 警 告

- Misuse of this product may lead to serious injury or death. To ensure the safety, make sure to read this manual and follow the instructions for assembly and setup carefully.
- Before operating, make sure the product is properly installed, or it may cause serious damage and injury.
- Make sure to disconnect the receiver battery firstly, then turning off the transmitter. If the receiver still been operated after transmitter is off, then it may lead to unintended operation and cause an accident.
- Before operating, ensure that all motors and servos are operated in the correct direction same with the transmitter transmitter. If not, make sure to trim the direction first.

- 遙控設備使用不當可能導致操作者或他人嚴重受傷、甚至死亡。為確保您和設備的安全，使用前請仔細閱讀使用說明書，並按照相關要求進行操作。
- 使用前必須確保本產品與模型安裝正確，否則可能將導致模型發生嚴重損壞。
- 關閉時，請務必先關閉接收器電源，然後再關閉遙控器。如果關閉遙控器電源時接收器仍然在運行工作，有可能導致控設備失控或是動力設備繼續運行，而引發不可預期的危險。
- 使用前，務必先確認模型所有伺服器的動作方向與操控遙桿方向一致，如果不同必須先調整好正確的方向，確認無誤，才能操作使用。

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CAUTION 注 意

IMPORTANT!

Always turn on the transmitter first, then the receiver. When turning off the system, always turn off the receiver first. The receiver should never be on without the transmitter on. Following this procedure will prevent damage to the servos and/or control surfaces. Or, in the case of electric-powered models, the motor may unexpectedly turn on causing severe injury.

重點提示！

每次開啓系統時，務必先開啓遙控器，再開啓接收器；當關閉系統時，則先關閉接收器，然後再關閉遙控器。接收器不應在遙控器未開啓前被開啓。按照以上程序操作可防止損壞伺服系統或控制舵面。否則在使用馬達動力的模型情況下，馬達可能會意外運轉造成嚴重傷害。

FORBIDDEN 禁 止 LOCATE AN APPROPRIATE LOCATION
遠離障礙物及人群

R/C aircraft can fly at high speed, thus posing a certain degree of potential danger. Choose a legal flying field consisting of flat, smooth ground without obstacles. Do not fly near buildings, high voltage cables, or trees to ensure the safety of yourself, others, and your model. Avoid location with magnetic and transmitter interferences. Please choose a legal flying field. Do not fly your model in inclement weather, such as rain, wind, snow or darkness.

操控遙控模型時具有一定的速度，相對的潛在著危險性，場地的選擇相當重要，請需遵守當地法規到合法的場地操作。必須注意周遭有沒有人、高樓、建築物、高壓電線、樹木等等，避免磁場干擾、外力訊號干擾及操控的不當造成自己與他人財產的損壞。請務必選擇在空曠合法專屬的場地。請勿在下雨、打雷、沙塵等惡劣天候下操作，以確保本身及機體的安全。



FORBIDDEN 禁 止 PROPER OPERATION
勿不當使用本產品

Do not attempt to modify the aircraft to alter its intended design. Please use only designated replacement parts listed in the manual to ensure its design structure integrity. Operate this product within its intended design parameters; do not overload it with excess cargo. This product is limited to personal hobby use, and pilot should be proficient with operation of this model. Follow all local law and ordinances when operating. Do not use this product for purposes which may violate others' personal privacy, and respect other's intellectual properties. Do not use this product for illegal purposes or beyond the bonds of common safety.

請勿自行改造加工，任何的升級改裝或維修，請使用亞拓產品目錄中的零件，以確保結構的安全。請確認於產品限界內操作，請勿過載使用，本產品為休閒娛樂專用之精密電子遙控飛行產品，僅限熟練遙控飛行器之個人使用，使用時請遵守當地法律規定，並嚴禁在任何違反公共安全區域操作，請勿利用本產品侵犯他人隱私 / 公開權、並遵重他人智慧財產權、著作權，且勿用於安全、法令外之其它非法用途。並充分了解您任何的使用與操作必須負完全的責任。



FORBIDDEN 禁 止 PREVENT MOISTURE
遠離潮濕環境

R/C aircraft are composed of many precision electrical components. It is critical to keep the model and associated equipment away from moisture and other contaminants. The introduction or exposure to water or moisture in any form can cause the model to malfunction resulting in loss of use, or a crash. Do not operate or expose to rain or moisture.

遙控模式內部也是由許多精密的電子零組件組成，所以必須絕對的防止潮濕或水氣，避免在浴室或雨天時使用，防止水氣進入機身內部而導致機件及電子零件故障而引發不可預期的意外！



A-TECH

WARNING 警告 DO NOT FLY ALONE
避免獨自操控

Before turning on your model and transmitter, check to make sure no one else is operating on the same frequency. Frequency interference can cause your model, or other models to crash. The guidance provided by an experienced pilot will be invaluable for the assembly, tuning, trimming, and actual first flight or unforeseen danger may happen. (Recommend you to practice with experienced pilots or with computer-based flight simulator firstly.)

操控模型前，需確認是否有相同頻率的同好正進行飛行，因為開啓相同頻率的遙控器將導致自己與他人立即干擾等意外危險。遙控模型操控技巧在學習初期有著一定的難度，要盡量避免獨自操作飛行，需有經驗的人士在旁指導，才可以操控執行，否則將可能造成不可預期的意外發生。(勤練電腦模擬器及老手在場指導是入門必要的選擇)




WARNING 警告 SAFE OPERATION
安全操作

Operate this unit within your ability. Do not fly while feeling impaired, as improper operation may result in danger. Never take your eyes off the model or leave it unattended while it is turned on. Immediately turn off the model and transmitter when you have landed the model.

請於自己能力範圍內並具備必要的操作技術的情況下操作遙控模型，過於疲勞、精神不佳或不當操作，意外發生風險將可能會提高。不可使遙控飛行機在視線範圍外飛行，降落後也請馬上關掉遙控飛行機和遙控器電源。




CAUTION 注意 ALWAYS BE AWARE OF THE ROTATING BLADES
遠離運轉中零件

During the operation of the multicopter, the rotor will be spinning at a high rate of speed. The blades are capable of inflicting serious bodily injury and damage to surrounding properties. Be conscious of your actions, and careful to keep your face, eyes, hands, and loose clothing away from the blades. Always fly the model a safe distance from yourself and others, as well as surrounding objects.

遙控飛行機主旋翼／螺旋槳是以高轉速運行，具有高危險性會造成自己與他人在身體上或環境上的嚴重損傷，請勿觸摸運轉中的主旋翼／螺旋槳，並保持安全距離以避免造成危險及損壞。




CAUTION 注意 KEEP AWAY FROM HEAT
遠離熱源

R/C aircraft are made of various forms of plastics, such as carbon fiber and polyethylene. Plastics are very susceptible to damage or deformation from extreme heat and cold climate. Make sure not to store the model near any source of heat such as oven or heater. It is best to store the model indoors, in a climate-controlled, room temperature environment.

遙控模型多半是以碳纖維、PA 纖維或聚乙烯、電子商品為主要材質，因此要盡量遠離熱源、日曬，以避免因高溫而變形甚至熔毀損壞的可能。





CAUTION 注意 SAFETY ON THE USE OF BATTERIES
電池使用安全

RISK OF EXPLOSION IF BATTERY IS REPLACED BY AN INCORRECT TYPE. DISPOSE OF USED BATTERIES ACCORDING TO THE INSTRUCTIONS.

如果使用不同型號的電池將可能有電池爆炸的風險。請依產品說明指示報廢電池。

XLGN

EQUIPMENT REQUIRED
自備設備



Tablet or Mobile Phone
平板電腦或手機

- The flight controller supports iOS V9.0 and higher ; Android V7.0 and higher.
- Considering the best working efficiency, the specifications of the tablet or mobile phone are recommended as follows: 4GB RAM, and more than 20GB ROM storage space; in order to keep the best and fast performance quality.

- 飛行控制器支援iOS版本9.0以上、Android版本7.0以上系統。
- 考量最佳使用效率，平板或手機的規格要求建議如下：搭配記憶體(RAM)4GB，以及儲存空間(ROM)20GB 以上；避免因平板或手機效能不足影響作業品質。

STANDARD EQUIPMENT
標準配備

 A13 GST Transmitter A13 GST遙控器	 Phone Screen Bracket 手機螢幕架	 A6B Receiver A6B接收器
 User Guidance 使用者須知	 Trainer Cable 教飛線	 Charging Cable 充電線

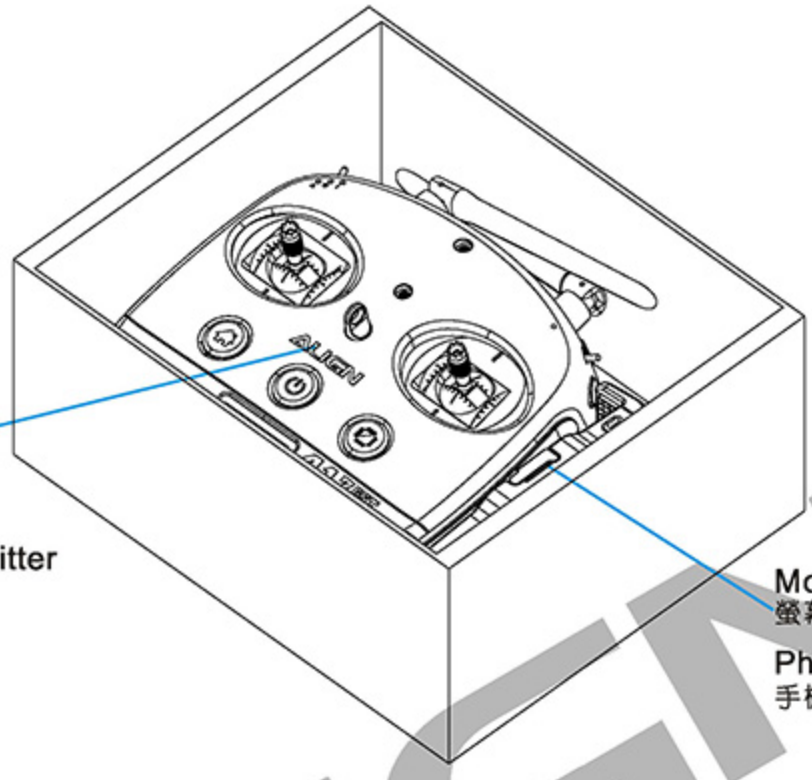
OPTIONAL EQUIPMENT
選配商品

 Simulator Cable FMS模擬線	 AP1 Multicopter Control Unit AP1多軸飛行控制系統	 AP2 Flight Control System Ap2飛行控制系統
 VTD1 1080 HD Digital Video Transmitter Set VTD1 1080高清數位圖傳組	 G2P Gimbal DV Set G2P雲台DV組	 1080 DV Camera 1080 DV鏡頭組x1

ALIGN

PACKAGE CONTENTS
包裝說明

ALIGN



Quick Finder
零件快速購

User Guidance
使用者須知

A6B Receiver
A6B接收器

A13 GST Transmitter
A13 GST遙控器

Option Equipment
選購品

Simulator Cable
FMS模擬線

Monitor Mount
螢幕固定座

Phone Screen Bracket
手機螢幕架

INTRODUCTION
產品介紹

ALIGN

ALIGN A13 GST transmitter and A6B receiver is a transmit/receive set can support upto 14 channels, and with 2.4GHz AFHDS digital proportional computerized R/C system which suitable for R/C aircraft. With AFHDS digital proportional computerized R/C system, anti-interference, low power consumption and high sensitivity receiver.

A13 GST transmitter is an ergonomic design with a comfortable grip and precision gimbal control. The built-in integration of this new remote control system was developed specifically to allow single button control of multiple functions and to improve flight efficiency.

Ground System APP is used to adjust the remote control parameters and custom button mode, so the aircraft maintains the best-performing state. Support for helicopter, multi-axis aircraft. Using a dual-frequency transmission system, with high-gain antenna, effectively enhancing the anti-jamming capability and picture quality.

ALIGN A13 GST遙控器和A6B接收器，為相對應的發射接收組合，遙控器具備14頻道作動，接收器最大支援可達14頻道。AFHDS是自動調頻數位系統的簡寫。 它是一個高度精密的遙控信號傳播系統， 這個系統能夠提供良好的距離，抗干擾能力強並且耗電量低。2.4G 增強版數位自動跳頻系統，適用於遙控模型飛行設備。具有數位自動跳頻系統，抗干擾、低功耗和高敏度接收。

A13 GST遙控器外觀符合人體工學設計，有舒適握感與精準靈敏搖桿控制，內在是完美整合性高全新遙控系統，而它就是為了高端專業產品而量身打造。可透過簡易按鍵控制飛行器多項功能，而提升操控效率。

支援APP調整遙控器參數以及自定義按鈕模式，讓飛行器保持最佳理想狀態。支援直昇機與多軸飛行器功能。採用雙頻發射系統，搭配高增益天線，有效大幅提升抗干擾能力及畫面品質。

SYSTEM FEATURES
系統特點

The AFHDS (Automatic Frequency Hopping Digital System) offering superior protection against interference while maintaining lower power consumption and high reliable receiver sensitivity. It is considered to be one of the leaders in the RC market today.

AFHDS 數位自動跳頻系統，是一個高度精密的遙控訊號傳播系統，具有抗干擾、低功耗和高敏度接收，是目前市面上最好的系統之一。



MULTI-CHANNEL HOPPING FREQUENCY 多頻段跳頻工作

This system bandwidth ranges from 2.408GHz to 2.475GHz, divided into 160 independent channels, each system using one of 16 different working frequencies and 160 kinds hopping regularity. It initiatives to avoid co-channel interference when power-up and using multi-channel hopping frequency. The 16 channels are selected at random from the list of 135 channels.

系統工作頻率範圍為2.408GHz至2.475GHz，共分為160個獨立頻點，每套系統使用其中不同的16個工作頻點和160種跳頻規律，通過開機時間不同及跳頻規律不同和使用頻點不同，主動避開同頻干擾。通信頻道是從135個頻道表中隨機選擇16個頻道跳頻。



OMNI-DIRECTIONAL GAIN ANTENNA 全角度增益天線

This system uses high-efficiency full-band gain antenna, which widely covers all the frequency band, it can greatly improve the efficiency of the system transmit and receive sensitivity, to enhance the system stability and passive anti-jamming capability.

此系統使用高效率的全波段增益天線，天線帶寬覆蓋整個波段寬度，可大大提高系統發射效率及接收的靈敏度，從而提升系統的穩定性，並增強自身被動抗干擾能力。



UNIQUE ID RECOGNITION SYSTEM 獨立身份識別系統

There is unique identification ID for each remote controller and receiver. After pairing with the receiver, this ID code will be stored in the receiver; when operating the receiver, you will be asked to verify this ID code. If verified failure, it's not activated. It improves and enhances system stability to prevent RF interference.

此系統每個遙控器和接收器都具有唯一的身份識別ID，當與接收器進行對頻後，此ID碼被存在接收器內；接收器工作時，首先會驗此ID碼，若驗證失敗則不會動作。此項可加大系統的主動抗干擾能力，進而提升系統的穩定性。



LOW POWER CONSUMPTION 低功耗

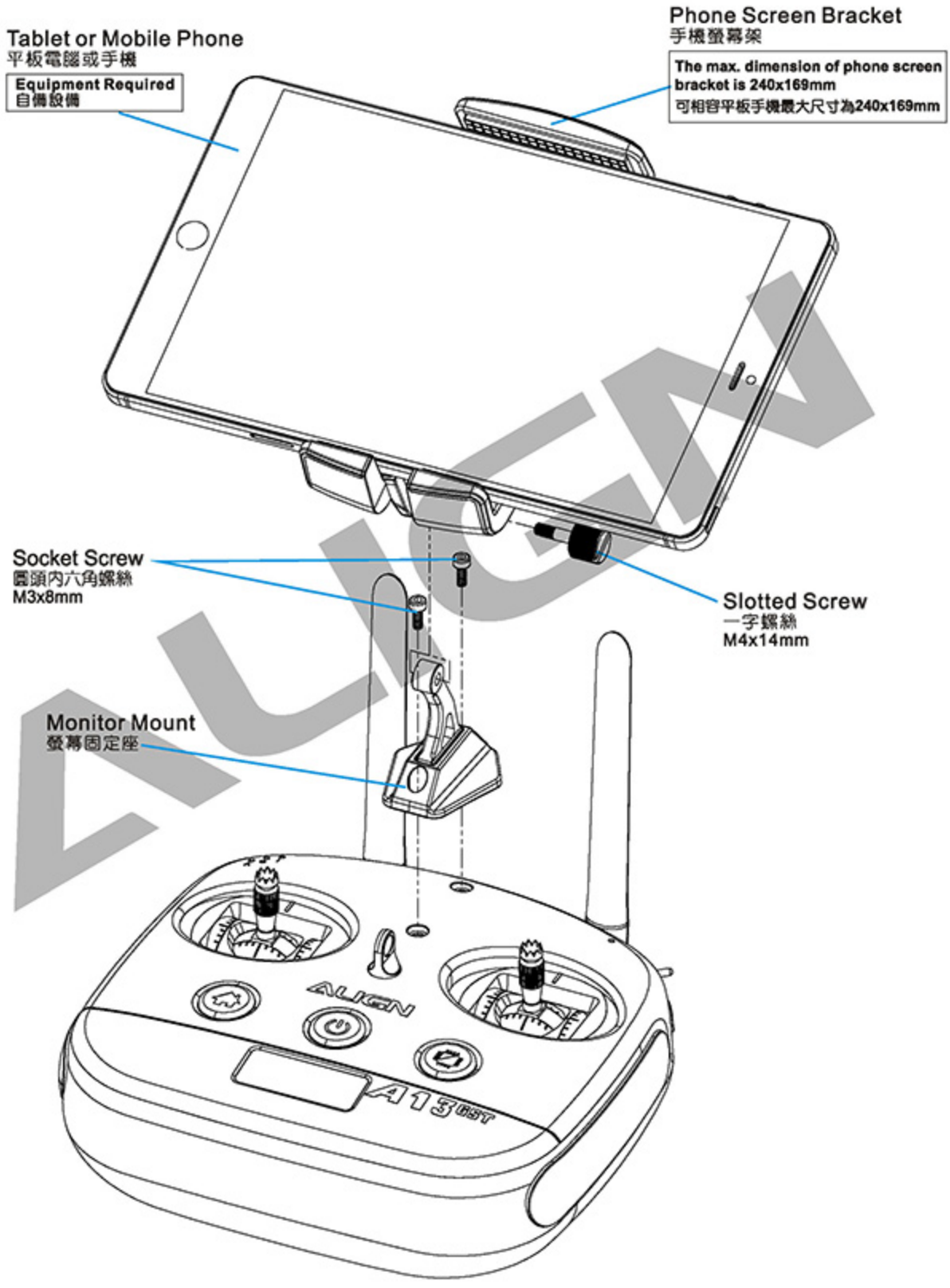
The system features with low power consumption, and highly sensitive component, while operating interval data transmission, it can effectively reduce the emission rate, extend battery lifetime, lower power consumption which is 1/10 of FM.

本系統採用低功耗、高靈敏度物件，同時使用間隔數據發送的工作方式，能有效降低發射率，延長電池使用時間，使系統功耗降低，是FM版本十分之一。

HELI

MONITOR MOUNT ASSEMBLY INTRODUCTION
螢幕架組裝示意圖

ALIGN

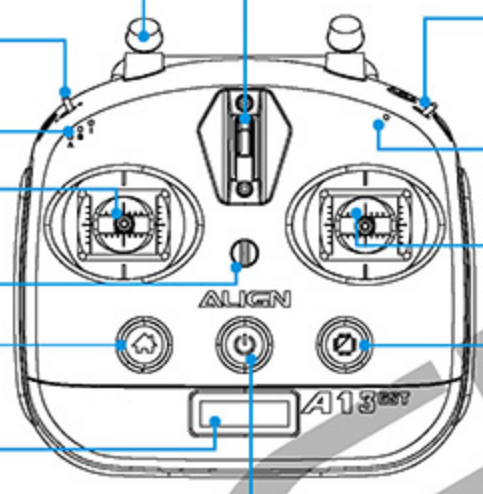


ALIGN

TRANSMITTER FUNCTIONS
遙控器功能介紹

ALIGN

1 TRANSMITTER INTRODUCTION
遙控器介紹



2.4G/5G Antenna
2.4G/5G 天線

Switch for Flight Mode(Left)
飛行模式開關(左)

Flight Mode LED Indicator
飛行模式指示燈

Stick (L)
搖桿(左)

Hatch Ring
吊環

Return Home (Press to trigger)
自動返航開關(按下即觸發)

LCD Panel
液晶顯示面板

Monitor Mount
螢幕固定座

Throttle Hold Switch (Right)
油門鎖定開關(右)

Self-Defining Function-B / Attitude Cruise Control LED Indicator
自定義功能-B / 姿態定速巡航指示燈

Stick (R)
搖桿(右)

ALIGN E1 : Self-Defining Function / Hook Switch / Trainer Mode Switch
ALIGN G1 : Self-Defining Function / Hook Switch / Choke Switch / Trainer Mode Switch
Multi-drone : Auto Retract Switch / Trainer Mode Switch
Helicopter : Self-defining / Trainer Mode Switch
ALIGN E1植保機：自定義功能 / 掛勾開關 / 教飛切換
ALIGN G1植保機：自定義功能 / 掛勾開關 / 阻風門開關 / 教飛切換
多軸機：自動收腳架開關 / 教飛切換
直昇機：自定義功能 / 教飛切換

Power Button
電源開關

Video Transmitter USB Port
圖傳USB插孔

Self-Defining Function-B / Attitude Cruise Control
自定義功能-B / 姿態定速巡航

Slider W1
Self-Defining Function-W1
滾輪開關/W1(自定義功能)

Grip
握把

Video Transmitter Control Box
圖傳控制盒

Agricultural Helicopter : Self-Defining Function - H / Auto Spray Switch
Multi-drone : Self-Defining Function - H / Hook Switch
Helicopter : Self-Defining Function - H
植保機：自定義功能-H / 自動噴灑開關
多軸機：自定義功能-H / 掛勾開關
直昇機：自定義功能-H

ALIGN E1: Photo / Video, AB Waypoint
ALIGN G1: Photo / Video, AB Waypoint, Engine Button
Multi-drone: Photo / Video
Short Press-Photo,
Extended Press-Video.
Helicopter: Photo / Video, AB Waypoint
ALIGN E1植保機： 拍照 / 錄影鍵、AB打點
ALIGN G1植保機：拍照 / 錄影鍵、AB打點、
引擎啓動鈕
多軸機：拍照 / 錄影鍵
短按-拍照 / 長按-錄影
直昇機：拍照 / 錄影鍵、AB打點

Slider W2
Gimbal Pitch Angle Control
滾輪開關/W2(雲台俯仰控制)

Handle Bar
提把

Cable Connection Port (Flip Cover)
接線孔蓋(掀蓋)



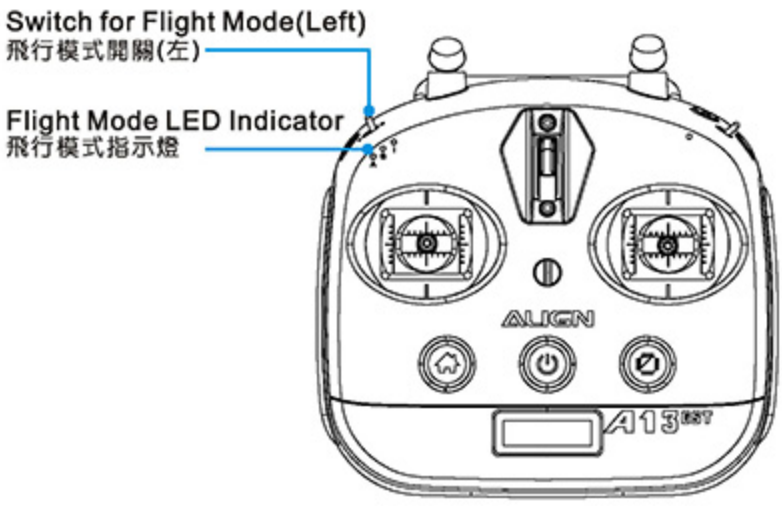
Charger Jack
充電插孔

Micro USB Port(Firmware Update)
Micro USB插孔(程式更新)

Trainer Port (Master Transmitter)
教飛線插孔(主從遙控器)

ALIGN

2 TRANSMITTER SWITCHES INTRODUCTION
遙控器開關功能說明



SWITCH FOR FLIGHT MODE INTRODUCTION 飛行模式開關功能說明



Stick move up : audio reminder / Flight mode LED "I" solid green light / Show "I" on the monitor. The modes shows in aircraft system :
Agricultural Helicopter : Auto Navigation Mode
Muticopter : GPS Velocity
Helicopter : Idle 2

撥桿往上，語音提示。飛行模式燈"I"綠燈恆亮。液晶顯示面板會出現"I"字樣。
植保機對應模式為：全自動導航模式
多軸機對應模式為：GPS(速度)
直昇機對應模式為：Idle 2



Stick move to middle : audio reminder / Flight mode LED "G" solid green light / Show "G" on the monitor. The modes shows in aircraft system :
Agricultural Helicopter : Manual GPS Mode
Muticopter : Attitude Mode
Helicopter : Idle 1

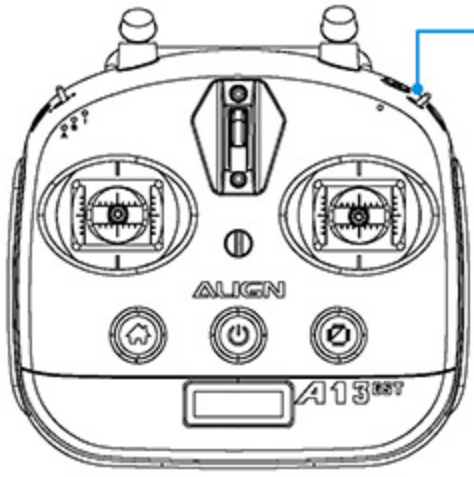
撥桿置中，語音提示。飛行模式燈"G"綠燈恆亮。液晶顯示面板會出現"G"字樣。
植保機對應模式為：手動GPS模式
多軸機對應模式為：姿態模式
直昇機對應模式為：Idle 1



Stick move down : audio reminder / Flight mode LED "A" solid green light / Show "A" on the monitor. The modes shows in aircraft system :
Agricultural Helicopter : Manual GPS Flight Mode (default setting is Manual Attitude Mode)
Muticopter : Manual Flight Mode (defult setting is Attitude Mode)
Helicopter : Normal

撥桿往下，語音提示。飛行模式燈"A"綠燈恆亮。液晶顯示面板會出現"A"字樣。
植保機對應模式為：手動GPS模式(出廠預設為手動姿態模式)
多軸機對應模式為：手動模式(出廠預設為姿態模式)
直昇機對應模式為：Normal

ALIGN



Throttle Hold Switch (Right)
油門鎖定開關(右)

Agricultural Helicopter : Throttle Hold Switch
Multicopter : Switch for Intelligent Mode
Helicopter : Throttle Hold Switch
植保機：油門鎖定開關
多軸機：智能模式開關
直昇機：油門鎖定開關

THROTTLE HOLD SWITCH INTRODUCTION 油門鎖定開關功能說明



Stick move down.
Agricultural Helicopter : Throttle hold, audio sound
Muticopter : Intelligent Flight Switch Off
Helicopter : Throttle hold release

撥桿往下。
植保機對應模式為：動力鎖定/熄火，語音提示。
多軸機對應模式為：智能飛行開關關閉
直昇機對應模式為：油門鎖定解除。



Stick move to middle
Agricultural Helicopter : on call, audio sound
Muticopter : Intelligent Flight Switch On
Helicopter : Throttle hold.

撥桿置中。
植保機對應模式為：動力待命，語音提示。
多軸機對應模式為：智能飛行開關開啓
直昇機對應模式為：動力鎖定。

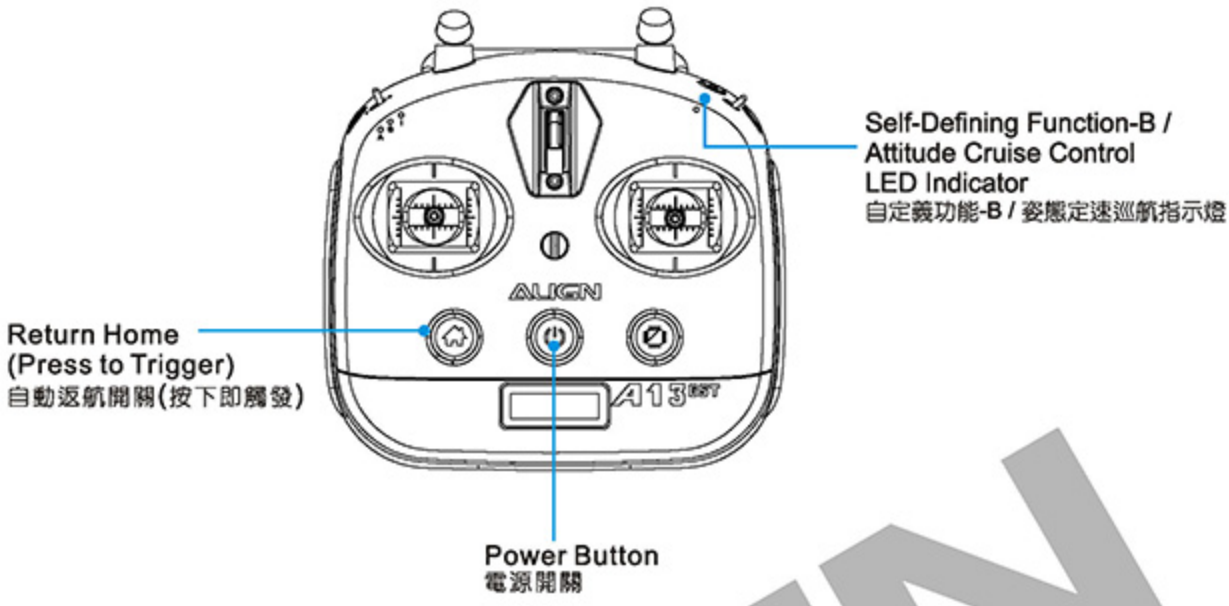


Stick move up
Agricultural Helicopter: stick move up and hold 2 sec. then back to center position to power release / throttle on
Helicopter : move stick to center position to throttle off

撥桿往上。
植保機對應模式為：推撥2秒回彈置中為動力解鎖/油門啓動。
直昇機對應模式為：推撥回彈置中為油門熄火。

ALIGN

3 TRANSMITTER BUTTON INTRODUCTION
遙控器按鈕功能說明



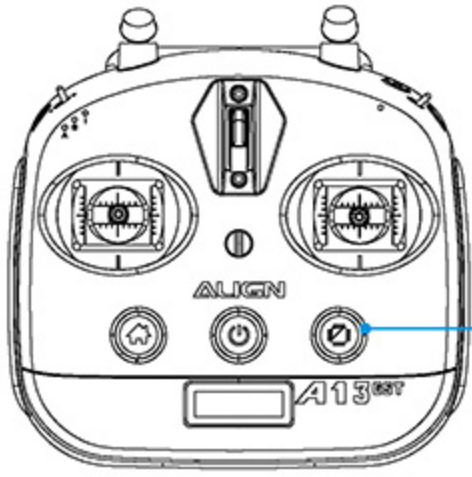
Auto Return Home Switch
Auto Return Home : Short-press to activate / Audio Reminder / Indicator LED solid white light. Disengage Auto Return Home : Short-press to disengage / Audio Reminder / Turn-off Indicator LED.

自動返航開關
自動返航：短按啓動，語音提示。指示燈白燈恆亮；
解除返航：短按解除，語音提示。指示燈關閉。

Power Switch
1) Power on : Extended-press 3 sec. / Audio reminder and alarm / Indicator LED white solid light / Turn on monitor screen.
2) Power off : Extended-press 3 sec. / Audio reminder and alarm / Indicator LED white solid light / Turn off monitor screen.
3) Program Version : Short-press 1 sec. / the latest program version will show on the monitor.

電源開關
1)開啓電源：長按電源鍵3秒，語音提示並發出開機聲響，指示燈白燈恆亮、液晶顯示面板開啓。
2)關閉電源：長按電源鍵3秒，語音提示並發出關機聲響，指示燈及液晶顯示面板關閉。
3)程式版本：短按，液晶顯示面板會出現目前遙控器程式版本。

ALIGN



ALIGN E1 : Self-Defining Function / Hook Switch / Trainer Mode Switch
ALIGN G1 : Self-Defining Function / Hook Switch / Choke Switch / Trainer Mode Switch
Multi-drone : Auto Retract Switch / Trainer Mode Switch
Helicopter : Self-defining / Trainer Mode Switch
ALIGN E1植保機：自定義功能 / 掛勾開關 / 教飛切換
ALIGN G1植保機：自定義功能 / 掛勾開關 / 阻風門開關 / 教飛切換
多軸機：自動收腳架開關 / 教飛切換
直昇機：自定義功能 / 教飛切換

1) Align E1 Function : Self-Defining Function / Hook Switch / Trainer Mode Switch (Connect with trainer cable)
- Hook Switch : After equipping the hook, short press hook switch
- Trainer Mode Switch : Connect with trainer cable, short press to start trainer function

Align G1 Function : Self-Defining Function / Hook Switch / Trainer Mode Switch (Connect with trainer cable) / Choke Switch
- Hook Switch : After equipping the hook, short press hook switch
- Trainer Mode Switch : Connect with trainer cable, short press to start trainer function
- Choke Switch : Short press to start chock switch

2) Multi-drone Function : Auto Retract Switch / Trainer Mode Switch (Connect with trainer cable)
- Auto Retract Switch: After equipping retract, short press to turn on auto retract switch
- Trainer Mode Switch : Connect with trainer cable, short press to start trainer function

3) Helicopter Function : Self-defining/Trainer Mode Switch / Trainer Mode Switch (Connect with trainer cable) Trainer Mode Switch : Connect with trainer cable, short press to start trainer function

1)ALIGN E1植保機對應功能：自定義功能 / 掛勾開關 / 教飛切換（需連接教飛線）。
- 掛勾開關：安裝掛勾組完成後，短按啓動掛勾開關。
- 教飛切換：連接教飛線後，短按啓動教飛功能。

ALIGN G1植保機對應功能：自定義功能 / 掛勾開關 / 教飛切換（需連接教飛線）/ 阻風門開關。
- 掛勾開關：安裝掛勾組完成後，短按啓動掛勾開關；
- 教飛切換：連接教飛線後，短按啓動教飛功能；
- 阻風門開關：短按啓動阻風門開關開關。

2)多軸機對應功能：自動收腳架開關 / 教飛切換（需連接教飛線）。
- 自動收腳架開關：安裝多軸起落架完成後，短按啓動自動收腳架開關。
- 教飛切換：連接教飛線後，短按啓動教飛功能。

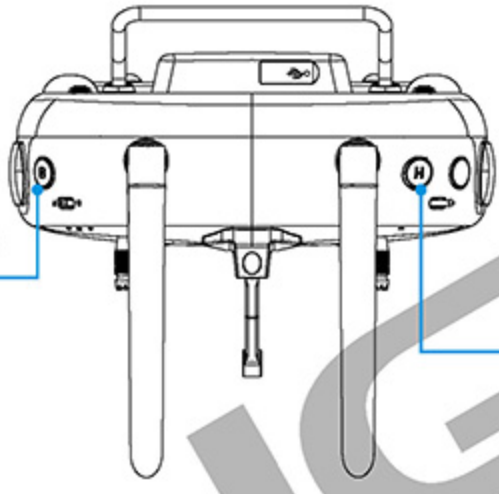
3)直昇機對應功能：自定義功能 / 教飛切換（需連接教飛線）。
教飛切換：連接教飛線後，短按啓動教飛功能。

ALIGN

CAUTION 注 意

To ensure quality of signal reception, the antenna should be vertical the RC model. When operating, careful to avoid pointing an antenna directly at the receiver. DO NOT grip transmitter antenna during operation as this will weaken the quality and intensity of transmitter transmission resulting in loss of control.

為確保訊號品質，天線應與遙控模型垂直放置。操控時不要用天線直接對著接收器。在使用過程中，嚴禁緊握遙控器天線，否將會減弱無線電傳播訊號的品質和強度，導致操控失靈及遙控模型失控的危險。



Self-Defining Function-B /
Attitude Cruise Control
自定義功能-B / 姿態定速巡航

Agricultural Helicopter : Self-Defining Function - H / Auto Spray Switch
Multi-drone : Self-Defining Function - H / Hook Switch
Helicopter : Self-Defining Function - H
植保機：自定義功能-H / 自動噴灑開關
多軸機：自定義功能-H / 掛勾開關
直昇機：自定義功能-H

B

Agricultural Helicopter : Self-Defining Function-B / Attitude Cruise Control
Multicopter Function : Self-Defining Function-B

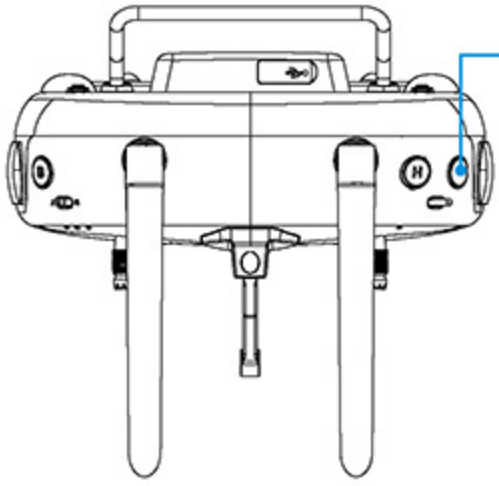
植保機對應功能：自定義功能-B / 姿態定速巡航
多軸機對應功能：自定義功能-B

H

Agricultural Helicopter : Auto Spray Switch.
Spray ON : Short-press to power on with audio sound and white LED solid light.
Spray OFF : Sort-press to power off with audio sound and LED light off.
Multi-drone : Self-Defining Function - H / Hook Switch
Helicopter : Self-Defining Function - H

植保機對應功能：自動噴灑開關。
啓動噴灑功能：短按啓動，語音提示，指示燈綠燈恆亮；
關閉噴灑功能：短按解除，語音提示，指示燈關閉。
多軸機：自定義功能-H / 掛勾開關
直昇機：自定義功能-H

ALIGN



ALIGN E1: Photo / Video, AB Waypoint
ALIGN G1: Photo / Video, AB Waypoint, Engine Button
Multi-drone: Photo / Video
Short Press-Photo,
Extended Press-Video.
Helicopter: Photo / Video, AB Waypoint
ALIGN E1植保機： 拍照 / 錄影鍵、AB打點
ALIGN G1植保機：拍照 / 錄影鍵、AB打點、引擎啓動鈕
多軸機：拍照 / 錄影鍵
短按-拍照 / 長按-錄影
直昇機：拍照 / 錄影鍵、AB打點

1)ALIGN E1: Self-defined Function、Photo / Video、AB Waypoint
ALIGN G1: Self-defined Function、Photo / Video、AB WaypointEngine Button. Extended Press until engine starts.
2)Multi-drone: Photo/Video
- Photo : Short-press to turn it on with audio sounds.
- Video Trigger : Press 2 sec. to turn it on with audio sounds.
- Video disengage : Press 2 sec. to shut it down with audio sounds. (For Aerial Photography Model

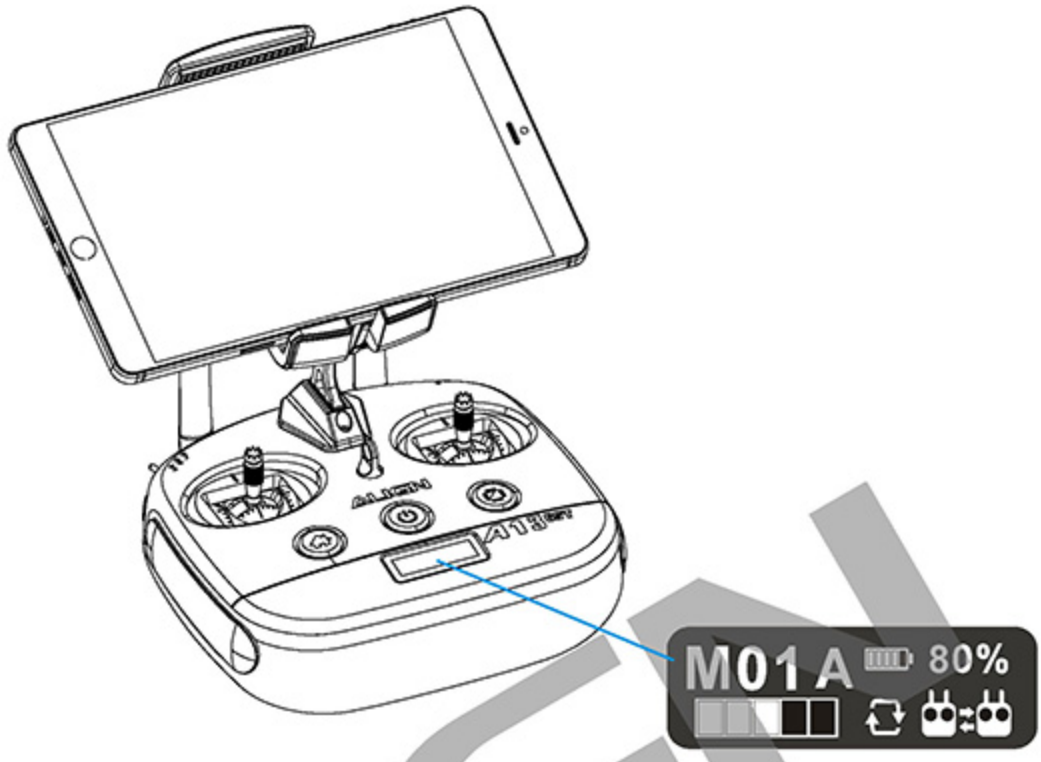
3)Helicopter: Photo / Video, AB Waypoint

1)ALIGN E1植保機對應功能：自定義功能、拍照 / 錄影鍵、AB打點。
ALIGN G1植保機對應功能：自定義功能、拍照 / 錄影鍵、AB打點、引擎啓動鈕。
引擎啓動鈕：長按直至引擎啓動即可。
2)多軸機對應功能：拍照/錄影功能。
- 拍照：短按啓動，語音提示。
- 錄影啓動：長按2秒啓動，語音提示。(　　)
- 錄影解除：長按2秒關閉，語音提示。(航拍機功能)

3)直昇機對應功能：自定義功能、拍照 / 錄影鍵、AB打點。

ALIGN

4 LCD PANEL ILLUSTRATION
遙控器液晶顯示面板圖示說明



LCD PANEL PICTURES 液晶顯示面板畫面	LCD PANEL PICTURES & ICONS ILLUSTRATION 液晶顯示面板畫面及符號圖示說明
M01	Model Selection 模型選擇。
A/G/I	Current Flight Mode : 顯示當前飛行模式
	Receive Signal (Under two bars, light will flash) 接收訊號，當接收訊號微弱低於兩格時，圖示閃爍提醒。
	Master / Slave trainer mode connected 教練機/主從機連線完成。
	Receiver Binding Completed 接收器對頻完成。
80%	Battery Power Indicator : The bar is flashing in electronic scroll when charging or flashing when attery power (in digital) under 20%. 電池電量指示燈：顯示當前電池容量，當電壓低於20%時，圖示閃爍、警示音提醒；充電時，圖示會漸進式閃爍
Binding...	Binding... 對頻中...
Binding ok	Binding ok 對頻完成。
Stick adjust...	Stick adjust... 搖桿校正中...
Stick adjust ok	Stick adjust ok 搖桿校正完成。
Reset Password?	Reset Password 重設密碼。

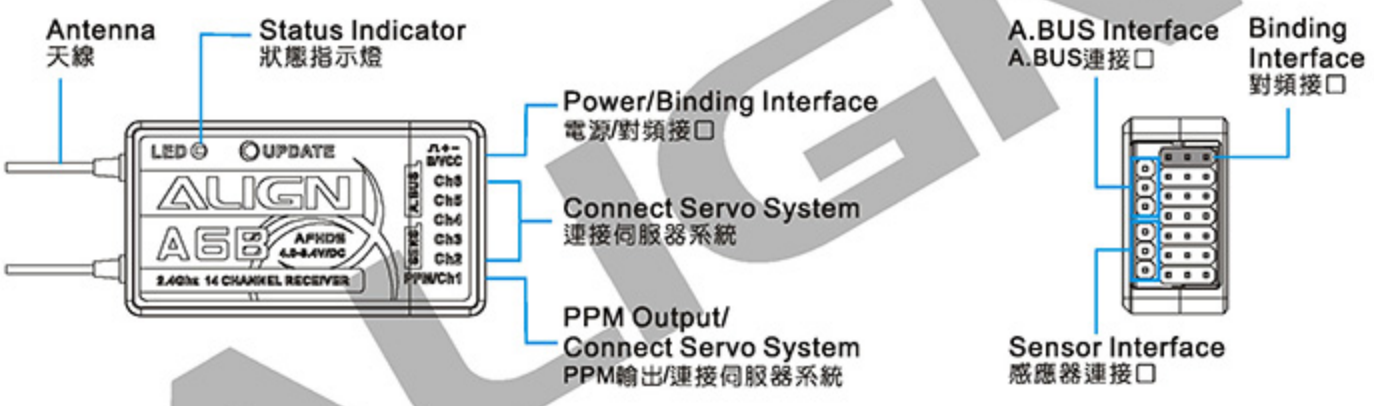
ALIGN

RECEIVER FEATURES
接收器特點介紹

A6B is bidirectional receiver especially for fixed Multicopter/ Wing / Gliders/ Helicopter, adopted AFHDSand omnidirectional antenna which has powerful anti-interference function. A6B supports 6 channels of PWM signal output and 14 channels of A.BUS signal and voltage detection interface. Also, A6B is provided with two receiver mode, divided 6 channels of PWM into CH1~CH6 or CH9~CH14 to make receiver more flexible and convenient in usage.

A6B是一款專用於多軸機/固定翼/滑翔翼/直昇機的雙向接收器，採用AFHDS並使用全向雙天線，具有超強的抗干擾功能。支援6個通道的PWM信號輸出與14個通道的A.BUS信號及電壓檢測接口。另具備兩種接收模式，可將6個PWM通道分配為CH1~CH6或者CH9~CH14，讓接收器的使用更加彈性方便。

1 RECEIVER SETUP INDICATORS
接收器各部位名稱



PPM/CH1 : Connect CH1 servo or PPM output signal.
CH2~CH6 :The interface connect servo, power and other accessories.
B/VCC : Using binding wire when binding; using power cord when normal operating, power range is 4.0~8.4V.
A.BUS: For output A.BUS signal.
SENS : For connecting sensor.

PPM/CH1：連接CH1通道的伺服器或輸出PPM信號
CH2~CH6：接口可以連接伺服器、電源或其他零件
B/VCC：對頻時用於連接對頻線，正常操作時用於連接電源線，供電範圍值在4.0~8.4V。
A.BUS：用於輸出A.BUS接收器接收訊號。
SENS：用於連接各感應器。

ALIGN

2 RECEIVER INSTALLATION
接收器安裝

CAUTION 注 意

IMPORTANT! In order to maximize the performance and enjoyment of the ALIGN A13 GST transmitter, please read this section carefully and completely. To ensure signal receiption quality, please make sure to keep electronics governor and other metal parts away when installing receiver.

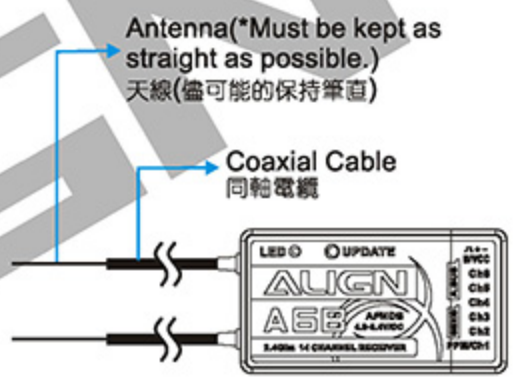
重點提示！為了將亞拓 A13 GST遙控器的功能發揮到最大並享受使用樂趣，請細心閱讀以下章節的所有內容。為確保訊號質量，安裝接收器時請盡量避開電子調速器及其他金屬零件。

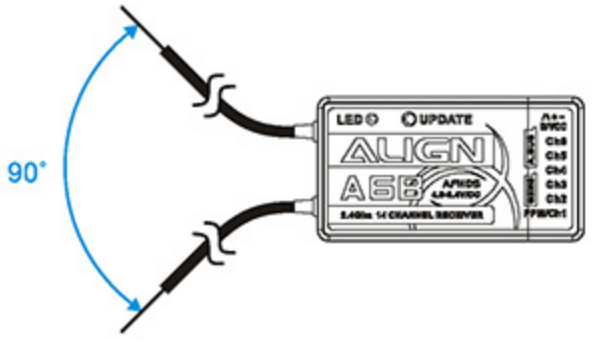
To obtain the best results from the diversity function, please refer to the following instructions:

要獲得多樣性功能的最佳效果，請參考以下說明：

The two antennas must be kept as straight as possible. Failure to do so might result in reduced operational range of the model.

1. Ideally, the two antennas should be placed at 90 degrees to each other. However, the most critical aspect is to keep the antennas away from each other as much as possible. Larger models can have large metal objects that can attenuate the RF signal. In this case the antennas should be placed at both sides of the model.
2. Then the best RF signal condition is obtained at any flying attitude. The antennas should be kept away from conductive materials, such as metal and carbon by at least halfinch. The coaxial part of the antennas does not need to follow these guidelines, but do not bend in a small radius.





兩個天線盡可能的保持筆直，否則將會導致模型的操作範圍降低。

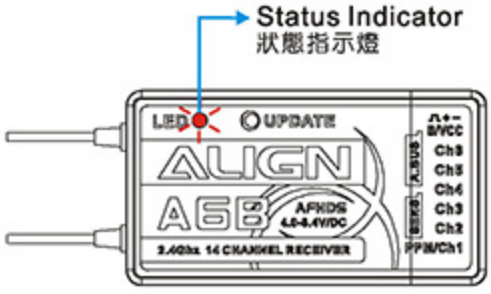
1. 理想的情況下，兩個天線放置應互為90度。最關鍵的環節就是保持兩個天線遠離彼此。較大的模型具有大型的金屬物體，容易衰減射頻信號。在這種情況下，強列建議將天線放置在模型的兩側。
2. 該天線應遠離導電材料如金屬和碳，至少半英吋的距離，藉以在任何條件飛行姿態下獲得最好的射頻信號。天線的同軸電纜部分並不需要遵循這些原則，但不要過度彎曲。



ALIGN

3 STATUS INDICATOR
狀態指示燈說明

The status indicator shows current battery voltage and work status of the receiver.
狀態指示燈用於指示接收器電源以及工作狀態



LED Off 燈號熄滅	Power disconnected. 接收器電源未連接
Red Solid Light 紅燈恆亮	Power on and working. 接收器已連接電源，並正常運作。
Fast Flashing 快速閃爍	Binding 接收器處於對頻狀態。
Flashing Slowly 慢速閃爍	The pairing transmitter is off or loss of signal. 已配對的遙控器未開機，或訊號遺失。

4 USAGE PRECAUTIONS
使用注意事項

1.After the linking is done, please cycle receiver power and check if the receiver to be linked is really under the control by the transmitter to be linked.
2.Please do not perform the linking procedure with motor's main wire is connected as it may result in serious injury.
3.To ensure signal receiption quality, please make sure to keep electronics governor and other metal parts away when installing receiver.

1.完成對頻後，請重開接收器電源，並確認接收器是否與遙控器完成正確對頻。
2.在對頻的過程中，請勿將馬達主電源接上，馬達可能會意外運轉造成嚴重傷害。
3.為確保訊號質量，安裝接收器時請盡量避開電子調速器及其他金屬零件。

ALIGN

GETTING STARTED
使用前準備

Before operating, make sure to follow the instruction to install the battery and connect the system.

開始操作前，請按照順序指引確認電池容量、連接設備。

TRANSMITTER CHARGING NOTICE 遙控器充電注意事項

Please follow below steps for transmitter battery checking :

1. Power on ALIGN A13 GST transmitter.
2. Check the battery indicator on the LCD monitor, please charge the battery when battery power under 20%.
3. Confirm the device is firmly commected.

請按照以下步驟檢查遙控器電池：

1. 開啓ALIGN A13 GST遙控器電源。
2. 檢查液晶顯示面板上的電池電量指示燈，若低於20%請先進行充電。
3. 檢查連接設備是否安插穩固。

WARNING 警告

- Ensure to double check battery indicator before every flight to prevent any unexpected danger happened.
- Ensure to use proper battery current for charging to prevent overcharging and cause damage. After full charge, please unplug the power cable to prevent overheat and fire or explosion.

- 每次飛行前，請再次確認遙控器之電池容量是否充足，電量不足可能導致不可預期的危險發生。
- 充電時，請檢查電池是否依適當的電流進行充電，避免因過度充電造成電池受損故障，電池充飽電後，請立即拔除電源，避免因長時間過熱引起火災發生，甚至更嚴重的爆炸事件。

OPERATION INSTRUCTIONS
操作指引

After finishing installation, please follow the instructions to start set up.

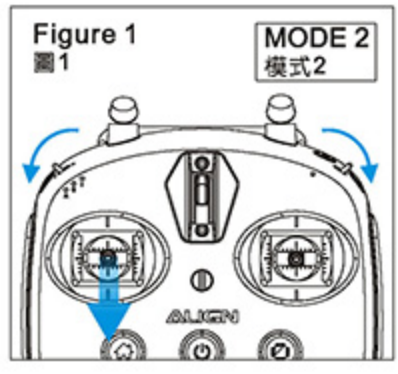
準備完成後，您可以按照本章指引開始使用本系統。

POWER ON 開機

Instruction for power on :

1. Check the system and make sure :
   a) The batteries are fully charged and installed properly.
   b) The receiver is installed correctly and disconnect the power.
   c) Put flight mode switch to "A" position; throttle hold switch stick move down, then put the throttle stick down. After power on, switch flight mode back to "G" position (Figure 1)
2. Slide the power button to the ON position. The screen will light up, and audible sounds indicating the transmitter is powered on.
3. Connect power supply to the receiver.

Figure 1 圖1



請按照以下步驟進行開機

1. 檢查系統狀態，確保:
   a)電池充滿電並且安裝正確。
   b)接收器安裝正確並且電源未連接。
   c)飛行模式開關指向"A"，油門鎖定開關撥桿往下並且將油門搖桿向下，開機完成後必須將飛行模式開關切回"G"(圖1)。
2. 開啓ALIGN A13 GST遙控器電源，直到螢幕亮起，此時正確開機會有開機聲響。
3. 連接接收器電源。

WARNING 警告

The system is now powered on. Use caution and always maintain a safe distance from the model. Serious injury may result if contact is made with the model during operation.

此時系統已啓動，請謹慎操作，並與模型保持安全距離，否則可能導致模型運轉而損壞，或造成人員傷亡、財務損失等，不可預期的危險發生。

ALIGN

TX / RX BINDING 遙控器/接收器對頻

The transmitter and receiver are bound in factory. Please follow the instructions if you use other transmitter and receiver :

1. Place transmitter & receiver within 1 meter range.

2. Power on transmitter. Two methods for RX binding: Binding via APP interface or press binding button on the transmitter.

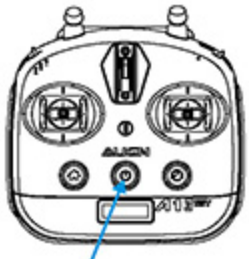
Binding Mothod I : Open A13 GST APP system, connect to bluetooth, press "Bind" botton on APP and power on RX to start binding. After binding completion, receiver red LED solid light.

本遙控器和接收器在出廠前已對頻成功。如果您要使用其他的遙控器及接收器，請按照以下步驟進行對頻：

1.遙控器與接收器放置在相互距離 1 公尺以內。

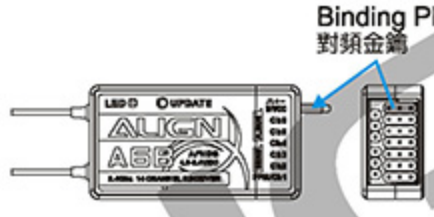
2.開啓遙控器電源，接收器對頻方式有兩種，透過APP對頻或按壓遙控器按鈕進行對頻。

對頻方式一：打開A13 GST遙控器APP，藍牙連線成功後，點按APP（對頻）進行接收器對頻，再開啓接收器電源進行對頻程序，對頻完成後，接收器紅色燈號恆亮。



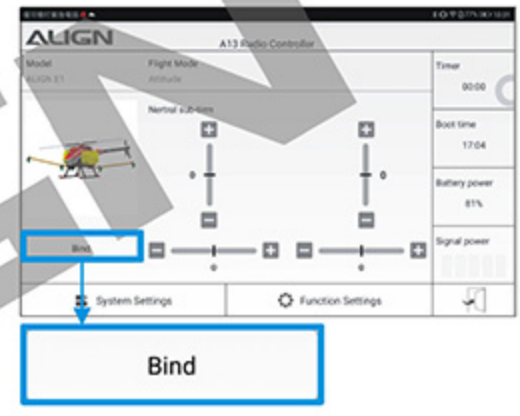
Power on A13

開啓遙控器電源



Receiver connect with binding Plug

接收器接上對頻金鑰。



Turn on APP and press "Bind" and transmitter screen displays "Binding..."

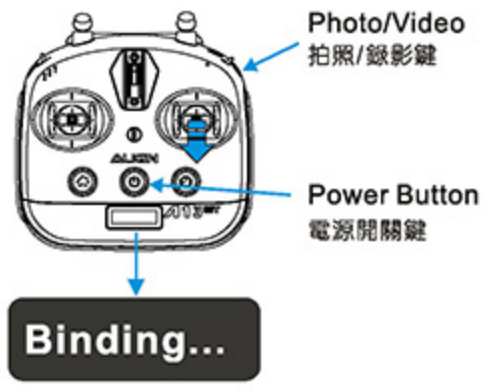
開啓APP連線後點擊（對頻），此時遙控器螢幕會顯示"Binding..."。

Binding Method II : Enter binding mode through hardware. Move the transmitter throttle stick down and press "Power button" and "Photo/Video button" at the same time, then power on RX to start binding. While binding completed, the RX red LED from flashing to solid light.

對頻方式二：可透過硬體進入對頻模式。將遙控器油門搖桿向下，同時按下遙控器（電源開關）及（拍照/錄影功能），再開啓接收器電源進行對頻程序，對頻完成後，接收器燈號會由紅燈閃爍轉變成紅燈恆亮。

Move the transmitter throttle stick down and press "Power button" and "Photo/Video button" at the same time, then power on RX to start binding. While binding completed, the RX red LED from flashing to solid light.

同時按下遙控器（電源開關）及（拍照/錄影功能），此時遙控器螢幕會顯示"Binding..."。



ALIGN

3. Receiver connect with binding key with power, A6B receiver get into process of binding.

4. Power on RX and complete binding, then RX LED red soild light.

5. Check transmitter, RX and model type works properly. Any question, please repeat binding process and above steps again.

3. 接收器接上對頻金鑰後給電，A6B接收器則自動進入對頻程序。

4. 接收器連接電源，並對頻成功正常運作，接收器燈號會顯示紅色恆亮。

5. 檢查遙控器、接收器、遙控模型是否正常工作。如有異常，請重複以上步驟重新對頻。



Power Button
啓動電源鍵



The monitor will show receiver signals and binding connection status after completion.

完成，此時遙控器上會顯示接收器訊號強度與連線對頻完成。



Solid red light on receiver after finish.

對頻成功後接收器燈號為紅燈恆亮

PRE-CHECK 操作前檢查

Before operating, please follow the instructions to check the system:

1. Check to confirm all systems are performing as expected.
2. Check reception distance for safety: the model is on the ground while one pilot observes the model. The other pilot while holding the transmitter walks away, increasing the distance of the transmitter from the model. Check the RC model status and mark the distance from where the model starts to lose control.

開始操作前，請執行以下步驟檢查系統狀態：

1. 檢查整個系統，確保各部位功能按照預期方式正常運行。
2. 按照如下方法確認安全距離：一名操作者操控遙控模型、另一操作者將手持遙控器帶向遠處，檢查遙控模型狀態，記錄遙控開始失效的距離。

CAUTION 注意

- When checking reception distance for safety, in order to avoid turning on the transmitter accidentally and causing RC model's blades rotated, suggested not equip blades on helicopter in advance; or directly remove the motor wiring temporarily reducing any unnecessary accident.
- Stop operating for any error condition.
- Make sure to operate the model within the safety receiving distance.
- Other singal interference may affect signal quality.
- 紀錄遙控模型有效安全距離時，避免因誤觸遙控器造成遙控模型主旋翼轉動，建議勿先將主旋翼安裝至機身上；或直接將機身上的馬達接線暫時移除可減少不必要的意外發生。
- 測試時，若有異常狀況出現，請不要操作遙控模型。
- 操作時，務必確保遙控模型未超出安全距離。
- 周圍干擾源可能會影響訊號的質量。

ALIGN

USB SIMULATOR　USB模擬器練習

Simulator function can simulate aircraft flight conditions on computer until familiar with the operation and then fly the real model.

1.Download FMS Simulator.

2.Open the FMS Simulator and connect transmitter and computer by USB.

3.Click Angalog control in Controls(Fgure 1), choose Joystick interface(Figure 2)，press Resourcesc then the transmitter information will show(Figure 3), press Mapping/Calibration can set up the channel and operate calibration(Figure 4). Press [OK] after setting to start simulator flight.

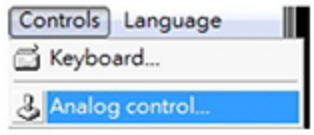


Figure 1
圖1

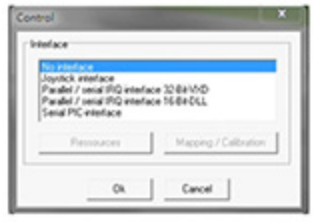


Figure 2
圖2

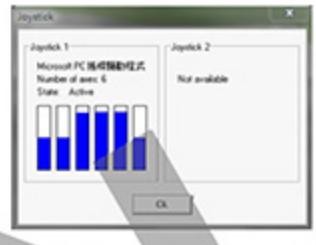


Figure 3
圖3



Figure 4
圖4

模擬器功能是指可於電腦上模擬搖控模型的飛行狀況，直到熟悉操作方式後再操控真正的模型。

1.請先下載FMS模擬器。

2.在電腦上開啓FMS模擬器，並將遙控器通過FMS模擬線連接到電腦上。

3.點擊Controls中的Angalog control(圖1)，選擇Joystick interface(圖2)，點下Resourcesc會顯示遙控器的訊息(圖3)，點下Mapping/Calibration可以設定頻道和進行校正(圖4)，全部設定完成點擊[OK]就可以開始模擬飛行了。

CAUTION 注 意

FMS Simulator Cable is optional accessary and not included in A13 GST transmitter package. You can refer to ALIGN website and search "Simulator Cable HEP00002T" for further purchase and practice.

A13 GST遙控器內無附"FMS模擬線"，若有需要使用FMS飛行模擬程式練習，請上亞拓官網選購專用的[HEP00002] FMS模擬線，即可使用FMS飛行模擬程式模擬練習。

POWER OFF　關機

Follow the instruction to turn off the system :

1. Disconnect the A6B receiver power.
2. Turn off the A13 GST transmitter .

請按照以下步驟進行關機：

1.關閉A6B接收器電源。

2.關閉A13 GST遙控器電源。

WARNING 警 告

Make sure to turn off the receiver power firstly, then turn off the transmitter to prevent any damage or serious injury.

關閉時，請務必先關閉接收器電源，再關閉遙控器，否則可能導致遙控模型損壞或造成人員受傷。

ALIGN

A13 GST TRANSMITTER SETUP
A13 GST 遙控器設定

A13 GST transmitter is flashed to the latest firmware version in factory.

Visit Align at www.align.com.tw for more news and firmware updates. Please scan QR Code for link to ALIGN website to find related software, or search" ALIGN A13" in Android app store.

A13 GST遙控器，在出廠前已是最新版本，請安心使用。

您也可以連結至亞拓官網查詢，隨時更新亞拓發佈的最新版本及各項最新訊息。請掃描QR Code連結亞拓網站下載相關軟體，或是在Android App store搜尋"ALIGN A13"。

http://www.align.com.tw/A13/

Compatible with ANDROID APP ON Google play

CAUTION 注 意

Recommend to use 8 inches laptop for system download.

建議使用8吋以上平板裝置下載。

1 PROGRAM SPLASH PAGE
程式版本宣告

Before enter to A13 GST transmitter home page, program splash page will display in 2 seconds.

The A13 GST transmitter Disclaimer is diplayed when the APP is launched for the first time. Please, read Align A13 GST transmitter Agreement in it's entirety. The first time you use the A13 GST transmitter, the system will display a disclaimer, please read the contents! By downloading, installing or using the ALIGN A13 GST Software or any part thereof, you agree to be complied with the terms and conditions.





第一次進入程式，或重新登入顯示版本宣告頁面兩秒後自動進入A13 GST遙控器首頁。

A13 GST遙控器免責聲明：
首次使用A13 GST遙控器，系統會顯示免責聲明，請詳讀內容！一旦下載、安裝或使用 ALIGN A13 GST軟體或其中任何部分，即表示貴用戶同意遵守各項條款與細則。

1 ENTER TO A13 TRANSMITTER SETUP HOME PAGE
進入A13 GST遙控器設定首頁

LANGUAGE 選擇語系

Support multi-language options. The system will automatically search and select proper language based on your laptop system. If your language is not in list, the default language interface is English. Or you can select a different language by clicking the icon at the uppet right corner.

支援多國語系，APP軟體會依下載的平板裝置使用語系自動判讀選擇。若無介面支援語系，APP軟體介面會切換成預設之英文介面。或可點按右上角圖示自行更改設定。



Language
選擇語系

SEARCHING BLUETOOTH DEVICE 搜尋藍牙裝置

Click〔Searching Bluetooth Device〕and choose〔ALIGN A13〕to start pairing. Enter your Bluetooth password and "Confirm". After pairing completion, A13 Transmitter screen will show the model (M01~M20), then A13 APP interface will show green solid light.

Bluetooth Light Status : ·Green Light : Connected.
·Red Light : Disconnected.

點按〔搜尋藍牙裝置〕，選擇〔ALIGN A13〕裝置進行連線。
連線後輸入藍牙密碼點按"確認"，藍牙連線成功後，A13 GST遙控器液晶顯示面板的會快速顯示M01~M20，APP面板模式燈號會亮為綠燈即代表連線成功。

連線狀態燈號說明：
・綠燈為已連線。
・紅燈為未連線。



CAUTION 注 意

Default Bluetooth Password is "00000000".

藍牙預設密碼為：00000000(8個零)。



ALIGN

CHANGE / RESET BLUETOOTH PASSWORD 修改 / 重設藍牙密碼

Click Change Bluetooth Password Key Icon for revise.

Please enter 1 ~ 8 number..

點按修改藍牙密碼圖示可變更設定。

請輸入1-8的字元數字密碼。



Change Bluetooth Password
修改藍牙密碼

CAUTION 注 意

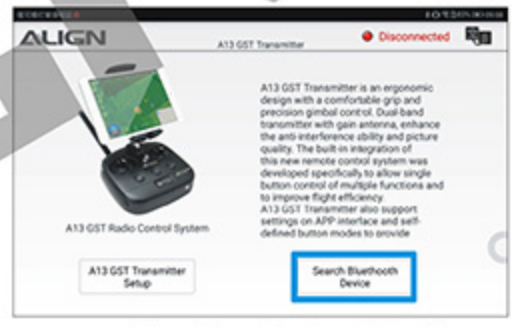
Strongly recommend to re-set your password to prevent interference from other devices.

強力建議使用者先更改設定密碼再使用，避免對其他藍牙裝置造成干擾。

If you forget your Bluetooth password, please click "Searching Bluetooth Device" to change password. Instruction as below:

1. Click"Searching Bluetooth Device" on A13 GST APP interface.
2. The A13 GST transmitter monitor screen will show "Reset Password?".
3. Then press"Auto Return Home" button on A13 GST transmitter to confirm for password reset. Or press"Power" button on transmitter to cancel password reset.

After password reset, please click"Searching Bluetooth Device" on APP interface again, then enter default password "00000000" for Bluetooth connection.



Searching Bluetooth Device
搜尋藍牙裝置

若忘記自行設定的藍牙密碼，點按搜尋藍牙裝置可變更設定。重設藍牙密碼步驟如下：

1.點按A13 GST APP上的"搜尋藍牙裝置"。

2.A13 GST遙控器液晶顯示面板會出現"Reset Password?"字樣。

3.點按A13 GST遙控器上的"自動返航開關"，確認重設藍牙密碼;點按A13 GST遙控器上的"電源開關"，取消重設藍牙密碼。

重設完成後，點按A13 GST APP上的"搜尋藍牙裝置"。使用原廠預設之藍牙密碼：00000000(8個零)，登入進行藍牙連線。



Default Setting
恢復原廠預設值



Auto Return Home Switch：confirm for password reset.
自動返航開關：確認重設藍牙密碼

Power Switch：cancel password reset
電源開關：取消重設藍牙密碼

ALIGN

After bluetooth connected, select〔A13 GST Transmitter Setup〕to A13 GST Setup Homepage. It displays the Home Page with valuable information including; model type, flight mode, neutral point adjustment, binding, timer, accumulated time, transmitter battery and signals...etc.

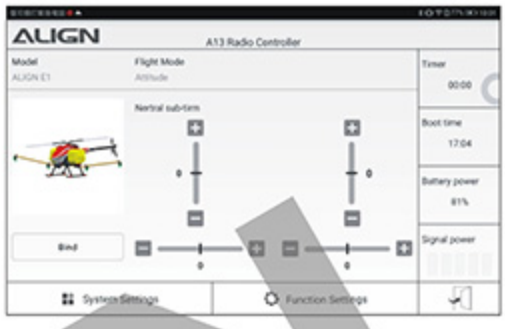
在藍牙已連線狀態下，選擇〔A13 GST 遙控器設定〕進入A13設定首頁。進入A13設定首頁後，顯示目前的模型顯示、飛行模式、中立點微調、對頻、計時器、累計時間、遙控器電量、訊號強度。

1)Connected, then click〔A13 GST Transmitter Setup〕to enter to Homepage.
已連線狀態，點按〔A13 GST 遙控器設定〕進入設定首頁。



A13 GST Transmitter Setup

Click〔A13 GST Transmitter Setup〕
點按〔A13 GST遙控器設定〕



A13 Homepage
A13 設定首頁

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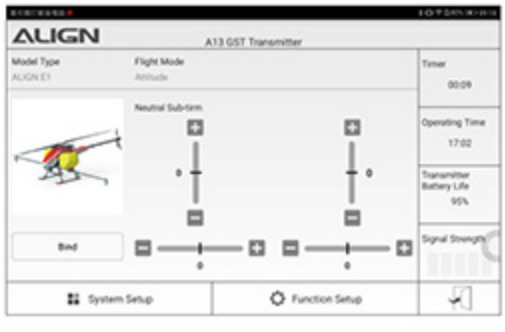
2)When system disconnected, click〔A13 GST Transmitter Setup〕, system will remind of Offline Mode, then enter to setup Homepage.
未連線狀態，點按〔A13 GST 遙控器設定〕後，會出現離線模式提醒再進入設定首頁。



A13 GST Transmitter Setup

Click〔A13 GST Transmitter Setup〕
點按〔A13 GST遙控器設定〕





A13 Homepage
A13 設定首頁

CAUTION 注意

When Bluetooth is disconnected in A13 GST APP, all parameter will show default settings and not allow for any adjustment.

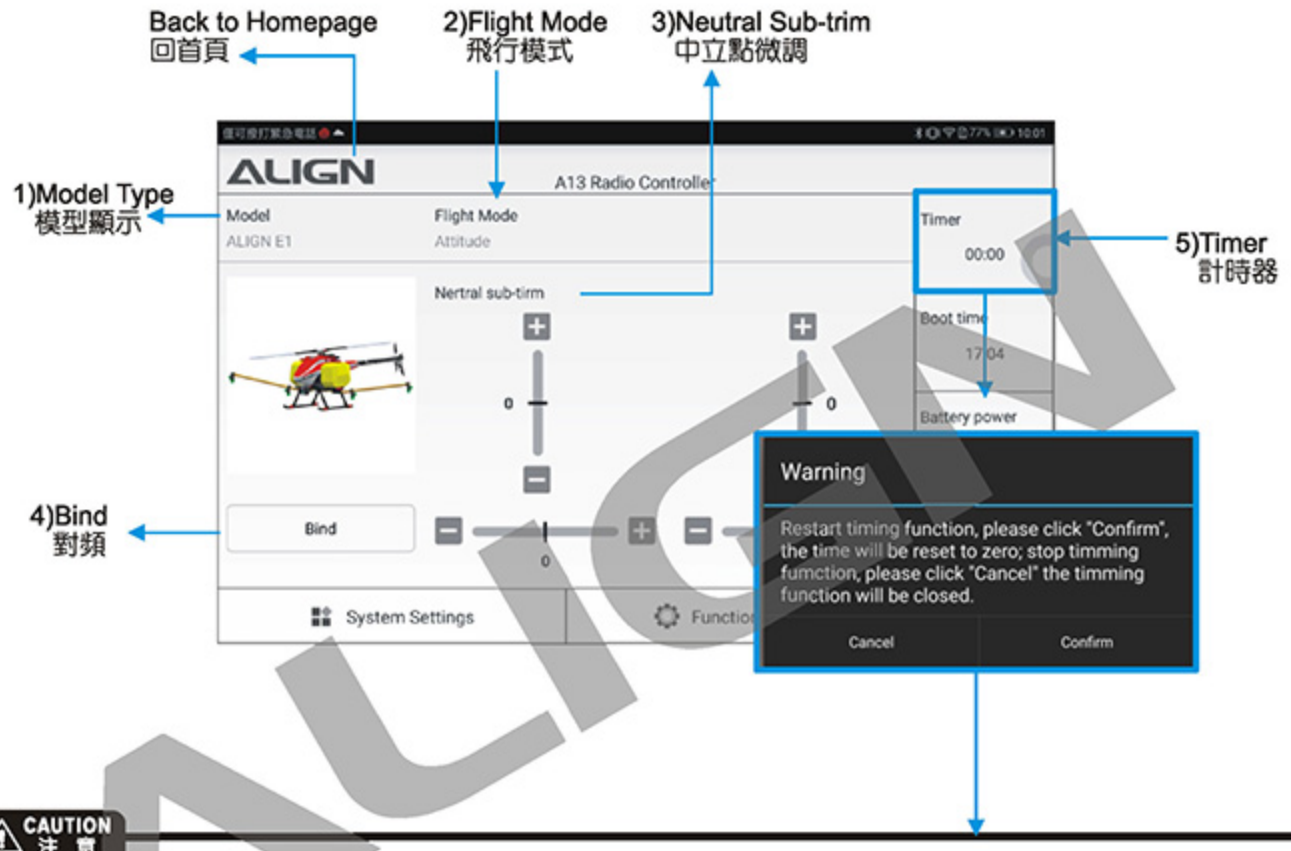
當A13 GST遙控器APP介面在藍牙"未連線"狀態下設定的參數皆無作用，所顯現的數據會依原廠預設值內容顯示。

ALIGN

2 A13 TRANSMITTER FEATURES INTRODUCTION
A13 GST遙控器設定功能介紹

Enter A13 GST homepage, it shows current model mode and flight mode. Also it provides hotkeys for "Bind" & "Timer". For more advanced settings, please click "System Setup" or "Function Setup" to enter the menu for details.

進入A13 GST設定首頁，顯示當前設置的模型模式及飛行模式，另外設定首頁也提供（對頻）及（計時器）的簡易設定快捷鍵，若需更進階的設定請點按（系統設定）或（功能設定）進入選單內設定其細項。



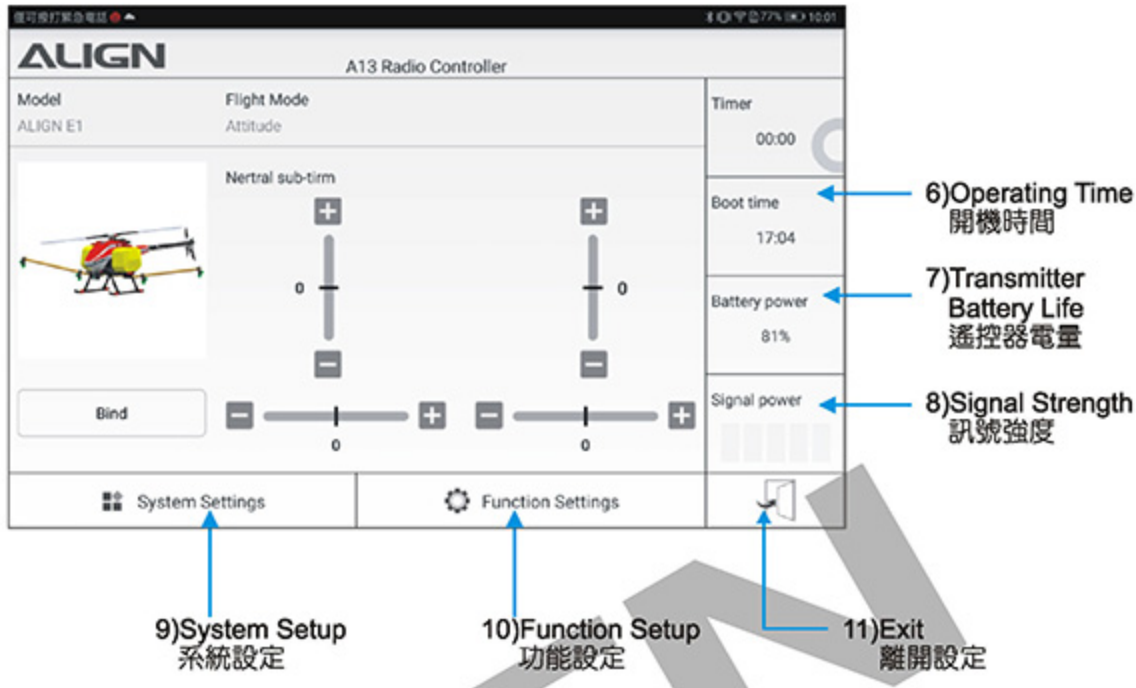
CAUTION 注 意

Restart timing function, please click "Confirm", the time will be reset to zero; stop timing function, please click "Cancel" the timing function will be closed.

重啓計時功能，請點按＂確定＂，時間將歸零重新開始；停止計時功能，請點按＂取消＂，關閉計時功能。

1) Model Type : Shows current model name, click and enter for model selection.
2) Flight Mode : Shows current flight mode.
3) Neutral Sub-trim: Allow to adjust stick center position of Elevator, Aileron, Throttle and Rudder by clicking "+ / -" botton gratually.
4) Bind : Click "Bind" to start transmitter binding process.
5) Timer : Shows current measure time.

1)模型顯示：顯示目前模型名稱，可進入模型選擇設定。
2)飛行模式：顯示當前飛行模式。
3)中立點微調：微調功能用來調整搖桿四個動作(升降、副翼、油門、尾舵)中立點的微調，可透過『＋、－』符號進行中立點的修正調整。
4)對頻：點按（對頻），遙控器即進行對頻模式。
5)計時器：顯示計時時間。



6) Operating Time : Shows Transmitter operating time.

7) Transmitter Battery Life : Shows transmitter battery life. If the voltage lower 20%, transmitter will make audible sound, please charge it immediately.

8) Signal Strength : Shows the signal strength of transmitter & receiver. The better signal, the more green LED will be on. Besides, ensure to check if the antenna is well-clear and in the receiption distance.

9) System Setup : Click and enter to system setup menu.

10) Function Setup : Click and enter to function setup menu.

11) Exit: Click and leave.

6)開機時間：顯示遙控器開機時間。

7)遙控器電量：顯示遙控器電量，當電壓低於20%時，遙控器會發出警示音，請立即為遙控器進行充電。

8)訊號強度：遙控器與接收器的訊號強弱顯示，訊號越強綠色燈號越多；反之，則請注意接收器的天線是否有遮蔽物阻擋或是已超出接收範圍。

9)系統設定：點按即可進入系統設定選單目錄。

10)功能設定：點按即可進入功能設定選單目錄。

11)離開設定：點按即可離開設定首頁。

2 SYSTEM SETUP
系統設定

Enter system setup menu, click or scroll down to enter system setup page.

進入系統選單，選擇想要的功能，點按或滑動畫面進入系統設定頁面。



1 MODEL SELECT
模型選擇

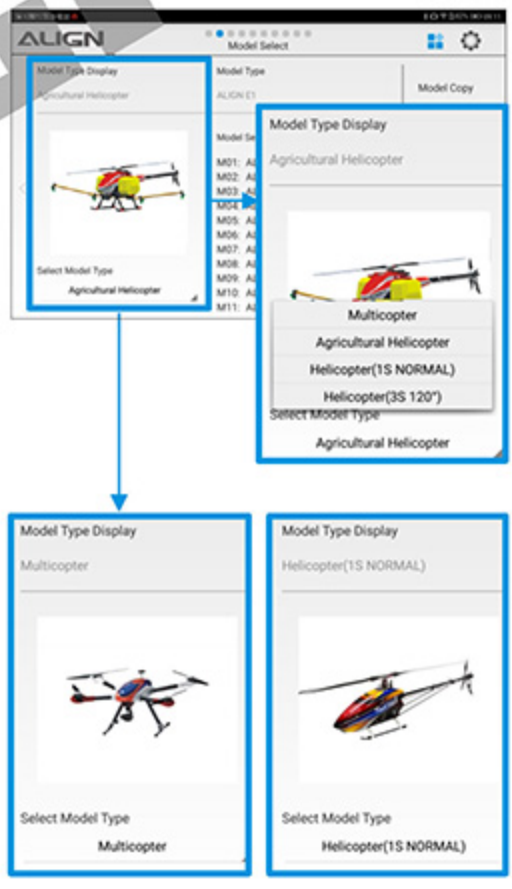
SELECT MODEL TYPE / MODEL TYPE DISPLAY　選擇模型類型 / 模型類型顯示

1) Select Model Type : A13 GST Transmitter provides 4 model types :
   1.Multicopter
   2.Agricultural Helicopter
   3.Helicopter (1S NORMAL)
   4.Helicopter (3S 120°)

2) Model Type Display : Default settings provides pictures of multicopter , agricultural helicopter and helicopter. Or it allows to change model picture based on your machine.

1)選擇模型類型：A13 GST遙控器提供四種模型類型選擇。
   1.多軸機
   2.植保機
   3.直昇機(1S NORMAL)
   4.直昇機(3S 120°)

2)模型類型顯示：顯示目前模型類型，原廠預設多軸機、植保機、直昇機三款圖片，也可點按圖片自行更換顯示模型之圖片。依選擇模型類型不同而有不同的圖示呈現。



ALIGN

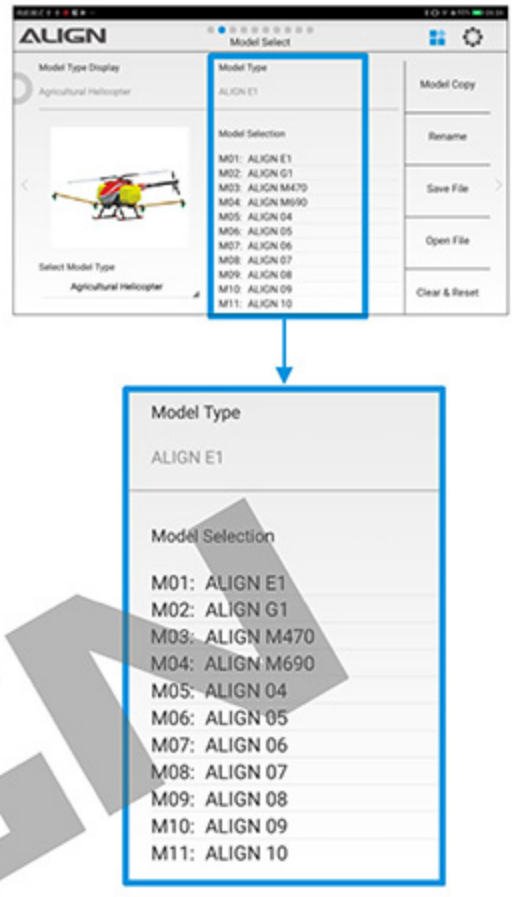
MODEL SELECTION / MODEL TYPE　選擇模型 / 模型顯示

1) Model Selection : Allow to save 20 sets of parameter settings and model types (M01~M20). Also allow to rename, copy, save or adjust the settings.

2) Model Type : Shows current model name and allow to rename it.

1)選擇模型：可用於記錄20組不同類型及參數的模型記憶(M01~M20)，可依個人喜好命名模型的名稱，進行命名、複製、儲存...等切換不同資料。

2)模型顯示：顯示目前模型名稱，依選擇模型名稱顯示。



CAUTION 注 意

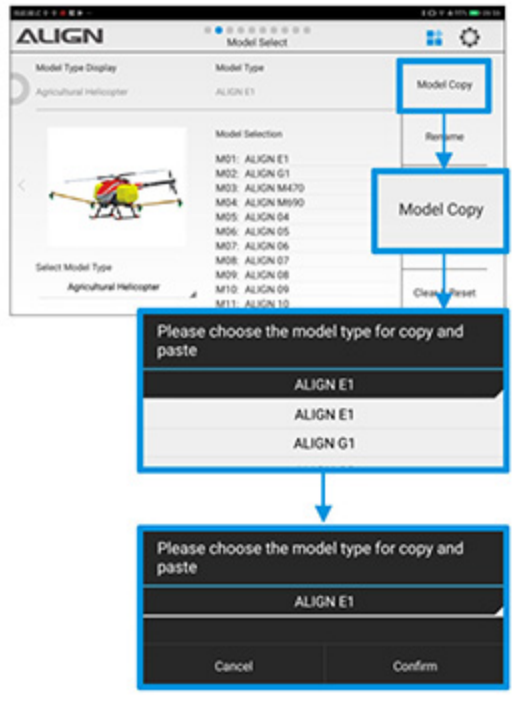
Model M01 refers "ALIGN E1".
Model M02 refers "ALIGN G1".
Model M03 refers "ALIGN M470".
Model M04 refers "ALIGN M690".
These four model type is default settings and not allow for changing.

預設模型M01 ： ALIGN E1 ： M02 ： ALIGN G1 ： M03 ： ALIGN M470 ： M04 ： ALIGN M690，此四組模型已綁定無法更改名稱。

MODEL COPY　複製模型

Select and copy model data to another folder. After copy and paste, the original data will be removed and replaced by the new settings.

將選定模型資料複製到目標模型中，目標模型的原始設置資料將被取代。



CAUTION 注 意

Ensure to double confirm the copy process as it cannot be restored.

複製完成後則無法回復原設定值，請務必再次確認是否執行此動作。

ALIGN

RENAME　重新命名

The name can be set with English and numbers, total 13 digitals based on your request.

名稱可由阿拉伯數字、英文大小寫及符號共13個字元組成，使用者可依據個人喜好命名模型的名稱，以利辨別。



SAVE FILE　儲存檔案

Save model parameter data in laptop device.

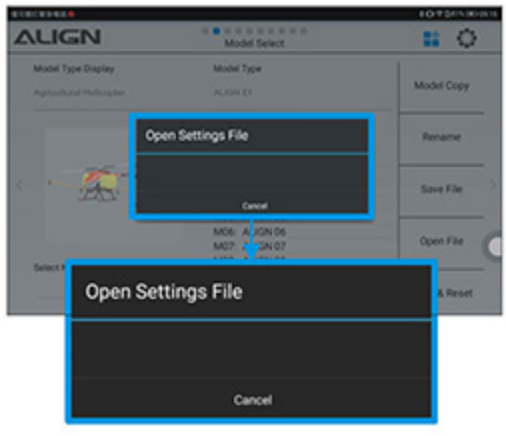
將模型參數檔案儲存至平板裝置上。



OPEN FILE　開啓檔案

Open and read model parameter data in laptop device.

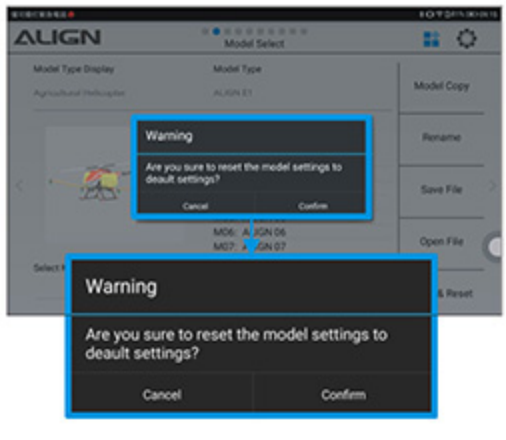
讀取平板裝置上的模型參數檔案。



CLEAR & RESET　清除模型

Clear current parameter and settings then reset to default setting.

清除當前選定的模型名稱及參數回復至原廠預設值。



ALIGN

2 TRANSMITTER SETTINGS
發射機設定

STICK MODE 搖桿模式

A13 GST transmitter support MODE 1 ~ MODE 4 stick modes. After selection, the stick mode and settings will change accordingly.

A13 GST遙控器支援MODE 1~MODE4共四種搖桿模式，設置完成後對應搖桿會自動切換，使用者可根據飛行習慣自行選擇。

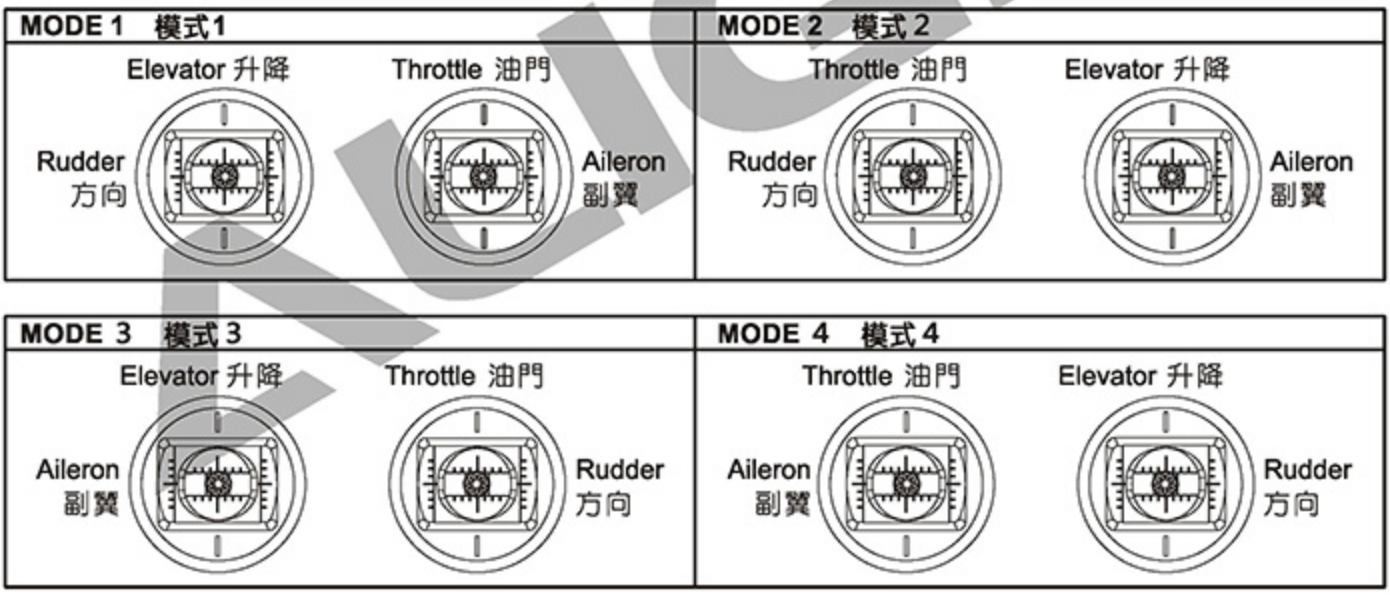


CAUTION 注 意

When changing the stick mode, please double check and perform a stick calibration to ensure all channels work correctly.

更換搖桿模式會改變遙控器操作習慣，更換完成後，請確認搖桿通道反應是否正確。





A13 GST TRANSMITTER MODE 1 AND MODE 2 CONVERSION INSTRUCTION
A13 GST 遙控器-模式1~2互換教學

Align also provides A13 GST transmitter Mode 1 and Mode 2 conversion instruction, please refer to following instruction video to convert it step by step to make sure your safety and proper use of this product. http://www.align.com.tw/A13/

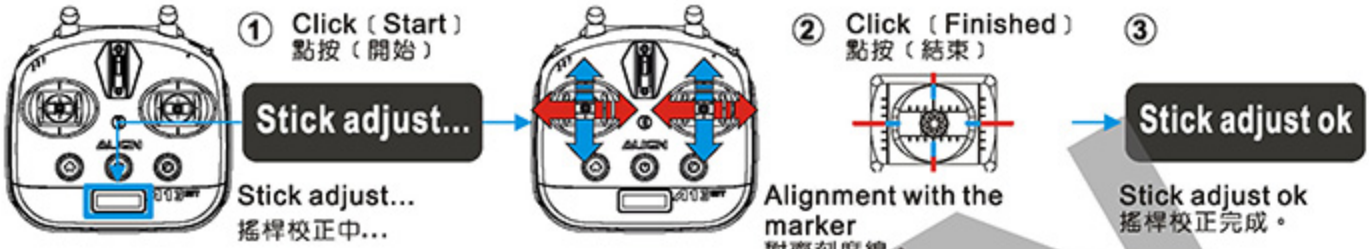
亞拓提供A13GST遙控器-模式1~2互換教學，請依循教學影片中的步驟拆解更換安裝，並利用下方連結處觀看，以確保您是安全且正確的使用本產品。網頁瀏覽 http://www.align.com.tw/A13

STICK CALIBRATION 搖桿校準

Adjust transmitter and slider neutral point and max. and min. endpoints. When enter to this page, click 〔Start〕 and move the sticks and sliders to the max. and min. stroke, then put them back to neutral position (alignment with the marker). Click 〔Finished〕for completion.

校正遙控器搖桿與滑軌開關的中立點與最大最小行程。進入此頁面後，點按〔開始〕將搖桿與滑軌開關打到最大最小行程後，回復至中立點位置(需對其刻度線)後，最終點按〔結束〕完成校正。





TRAINER MODE 教練模式

Suitable Models： Helicopter & Agricultural Helicopter
適用模型類型：直昇機、植保機

The feature is only suitable for helicopter and agricultrual helicopter setup.

The mode status will be varied from different model types. Connection status will be shown as below. Trainer Mode student can make two transmitters in connection and control at the same time through trainer cable. One transmitter is controlled by instructor while the other one by student. When trainer mode is on, instructor can control the model on his own transmitter and channels definition is locked, students can not control any model. When trainer mode is off, student can control by themselve freely.



此功能適用於直昇機、植保機模型選擇。

依使用者選擇之模型類型不同，而出現對應之模式狀態，下方會出現連線狀況。初學者可透過教飛線連接兩台遙控器一同控制。一台遙控器為教練控制，另一台遙控器為初學者學生控制。教練可通過自己的遙控器控制模型。當教練關閉開關時，初學者則恢復控制權。學員機上所有資料都被鎖定，操縱杆的資料直接由教練機開放通道定義，學員機無法直接控制任何模型。

CAUTION 注 意

When trainer mode is on, any other transmitter connected to instructor transmitter should be off. All functions on student's transmitter is locked to prevent any flight damage or signal interference.

任何與教練遙控器連線的遙控器必須關閉。學員機上所有的設置資料將被鎖定，教練機和學員機共用教練機上的設置資料防止教練開關打開時教練機和學員機的脈衝干擾。

ALIGN

MASTER / SLAVE MODE　主從模式

Suitable Models： Multicopter
適用模型類型：多軸機

This feature is suitable for Multicopter setup. The model type will be varied from different model types. Connection status will be shown as below.

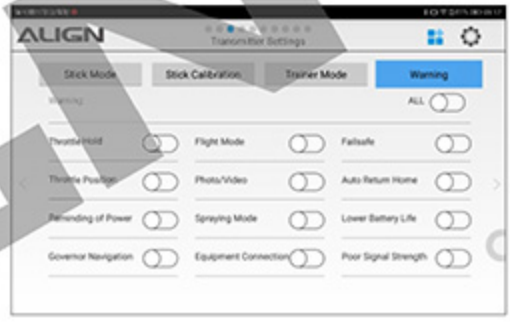
此功能適用於多軸機模型選擇。依使用者選擇之模型類型不同，而出現對應之模式狀態，下方會出現連線狀況。



WARNING　警示提醒

For transmitter warning notice setup. Allow to make it on or off personally. For audible sound, please go to〔Volume Setup〕for adjustment.

設置發射機的警示提醒功能。可依照個人使用需求開啓或關閉警示提醒。警示音量請至〔音量調整〕中設定警示音量大小。



3 FAILSAFE SETTINGS
失控保護設定

A13 GST transmitter failsafe provides rescue protection. When receiver lost signal, the server or flybarless system will return to default setting. Failsafe provides two protection setup:

1. When failsafe is "OFF", server will keep at current position before lost signal.
2. When failsafe is "ON", server will perform to rescue position.

失控保護設定A13 GST遙控器提供之 安全保護功能，當接收器失去訊號或中斷時，伺服器或飛控系統將恢復至預設值位置。

失控保護設定提供兩種保護設定：

1. 保護設定值為"OFF"時，則保留伺服器中斷信號前的最後一個動作指令。
2. 保護設定為所設置的保護位置時，當中斷信號時就會執行設置的保護位置指令。



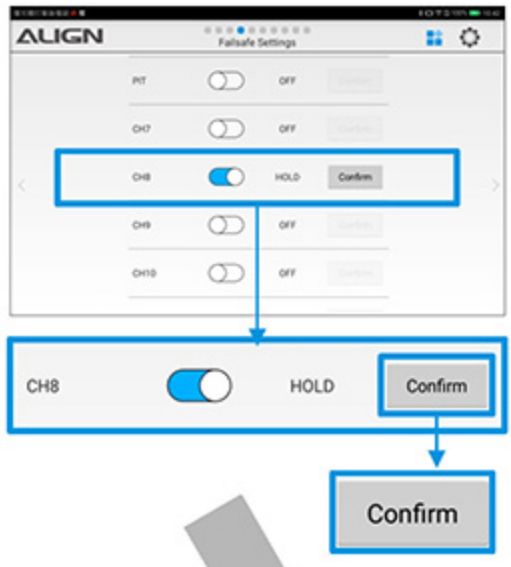
ALIGN

FAILSAFE PROTECTION SETTING INSTRUCTION

失控保護設定步驟說明

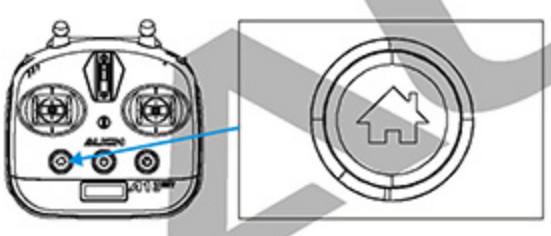
1.Please turn on failsafe protection setting (CH8) firstly, and the data will become HOLD after turning on.

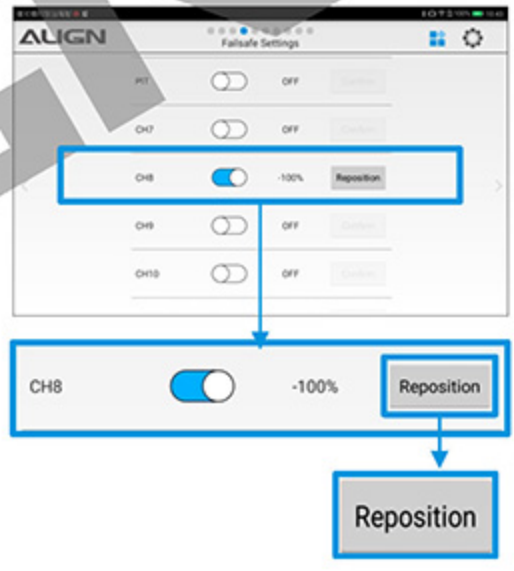
1.請先開啓失控保護設定(CH8)，開啓後數據變成HOLD。



2. Click the Confirm button and the data will change from HOLD to -100%. Click the Reposition button, and clicks the Auto Return Home button on A13 GST transmitter.

2.點按確認鍵，數據會由HOLD變成-100%。再點按復位鍵，此時A13 GST遙控器點按自動返航鍵。





3. Click the Confirm button, the data will change from -100% to 100%, that is, complete the fail protection setting step.

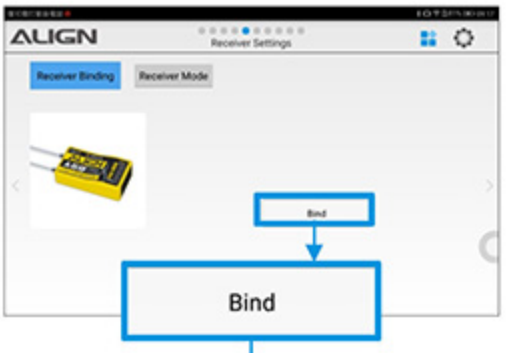
3.點按確認鍵，數據會由-100%變成100%，即完成失控保護設定步驟。



RECEIVER BINDING 接收器對頻

Receiver Binding: Select receiver type then process binding.

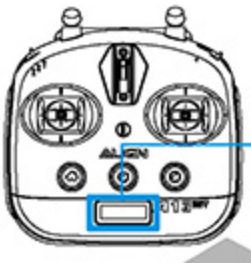
接收器對頻：選擇接收器類型進行對頻。點按（對頻）鍵後會跳出"是否進行對頻"的警示窗。



Click"Confirm", APP will back to A13 setup home page and the transmitter monitor and A13 GST setup homepage will show "Binding...".

點按（確認）後，畫面會自動跳回A13設定首頁，此時遙控器螢幕及A13 GST 設定首頁顯示會顯示"Binding..."。





① Binding...

"Binding..."
對頻中

② Binding ok

"Binding ok"
對頻完成

CAUTION 注意

All functions will stop, please do not turn off APP and transmitter power or do any setup.

所有功能將暫停，請勿關閉APP及遙控器電源或做任何動作設定。



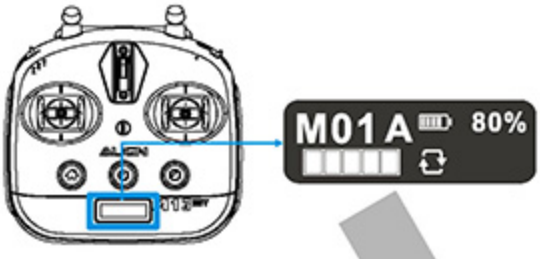
If it takes over 1 mins in binding, then system will show "Binding fail. Please rebind." Ensure to check if the antenna is well-clear and in the receiption distance. Re-click（Bind）botton on APP and power on RX to start binding.

對頻時間若超過1分鐘會自動退出顯示對頻失敗。請注意接收器的天線是否有遮蔽物阻擋或是已超出接收範圍。繼續點按APP（對頻）進行接收器對頻。



Receiver will show solid red LED after binding completion. On transmitter monitor and A13 GST setup homepage, the signal strength bar will light up green. The more refers to stronger signal strength.

對頻完成後，接收器紅色燈號恆亮。遙控器螢幕及A13 GST 設定首頁顯示訊號強度越強，則燈號越多。

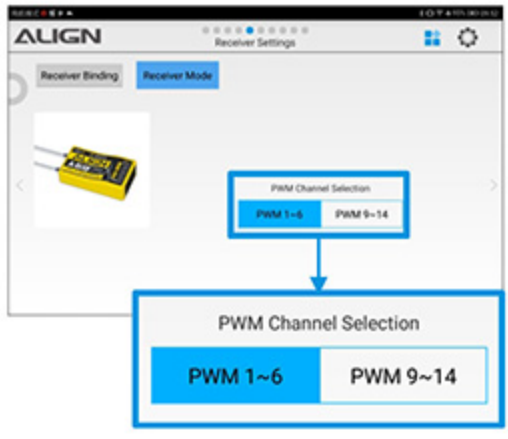




RECEIVER MODE 接收器模式

Receiver Mode : PWM Channel Selection
Using ALIGN A6B receiver, two receiver modes for selection:

Mode 1 : Set the receiver CH1 ~ CH6 sockets correspond to the channel 1 ~ channel 6 output.
Mode 2 : Set the receiver CH1 ~ CH6 socket corresponds to the channel 9 ~ channel 14 output. You can set the A6B receiver mode personally.



接收器模式：PWM通道選擇

搭配ALIGN A6B接收器使用，可選擇A6B的兩種接收模式。

模式1：接收器CH1~CH6插座對應頻道1~頻道6的輸出：

模式2：接收器CH1~CH6插座對應頻道9~頻道14的輸出。可依使用者所需來選擇A6B接收器的接收模式。

5 TIMER SETTINGS
計時器設定

Two ways of timer setting : Setup a countdown or start counting from "0". Pull the throttle up then start counting. You can setup your own timer personally.

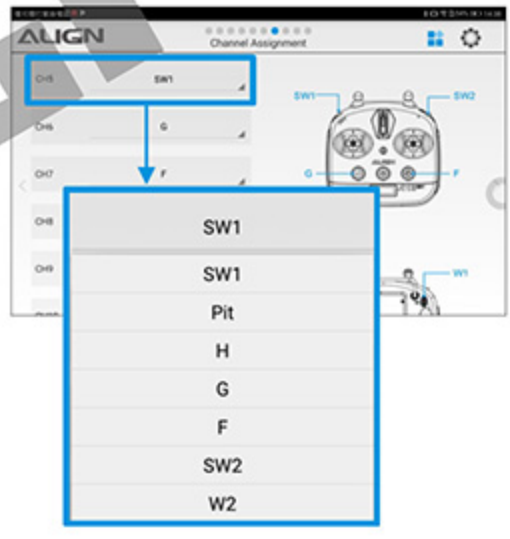
計時器設定分為倒數計時或正數計時，推啓油門即啓動計時開關。使用者可依操作習慣設定開始開關。



6 CHANNEL ASSIGNMENT
通道開關設定

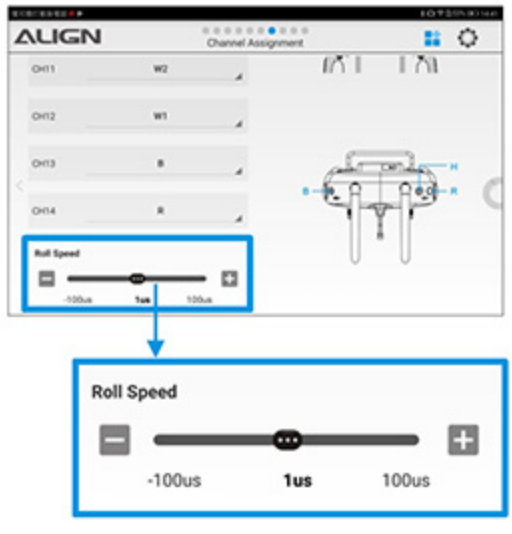
Set or revise of CH5~CH14 channel switch according to user's requirement.

依使用者需求設定或修改CH5~CH14通道位置的控制開關。



Allow to set up gimbal rolling speed individually. The bigger number refers to faster rolling speed.

依使用者需求設定雲台滾轉速度的調整量，數字越大滾轉速度幅度越大。

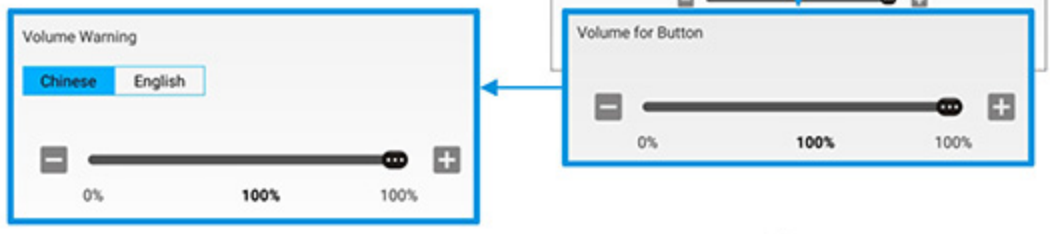


ALIGN

7 VOLUME SETUP
音量調整

Adjust the volume of transmitter button, warning and select voice language in Chinese or English.

可調整遙控器按鈕音量、警示語音音量及選擇警示語音，警世語音提供中文與英語兩種選擇



8 RESET TO DEFAULT SETTINGS
恢復原廠預設值

Allow restoring parameters to default settings when A13 GST Transmitter needs to be reset.

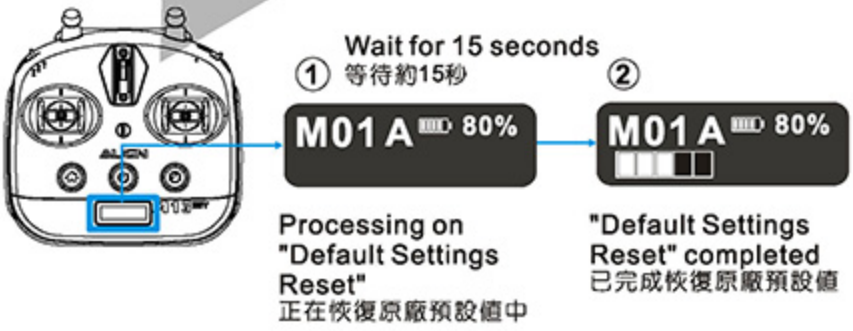
After clicking "Confirm" button, it will start processing while the signal receiption bar disapears. Wait around 15 seconds, the signal receiption bar will show up again after the process is completed.

當A13 GST遙控器欲重新設定時，可利用此功能將遙控器所有參數恢復至原廠預設值。

點按"確認"鍵後，A13 GST遙控器液晶面板上的接收訊號現圖示會消失，代表正在恢復原廠預設值中。

等待約15秒，A13 GST遙控器液晶面板上的接收訊號會再次出現圖示，代表已完成恢復原廠預設值。





CAUTION 注 意

Reset parameter to default setting, all the settings will be deleted and set to factory setting accordingly.

參數恢復至原廠預設值後您所有設置的參數將會遺失。

ALIGN

9 FIRMWARE UPDATE
程式更新

FIRMWARE DOWNLOAD 主程式下載

Current transmitter firmware version and update date will be shwn in the upper right corner. Turn on mobile or Wi-Fi and click〔firmware download〕to download it.

右上角會顯示目前遙控器主程式版本及更新日期。請先開啓行動網路或Wi-Fi點按〔主程式下載〕下載主程式。



CAUTION 注 意

All functions stop while undating firmware, do not turn off APP and transmitter power or do any setup or it might cause damage of A13 transmitter.

主程式版本更新時，所有功能將暫停，請勿關閉APP及遙控器電源或做任何動作設定，否則可能會造成A13遙控器當機損壞。

FIRMWARE UPDATE 主程式更新

After downloading, click〔firmware update〕and update process will be shown accodingly.

下載完成後，點按〔主程式更新〕進行更新，下方會出現更新進度。



CAUTION 注 意

After update finish, A13 GST transmitter will auto power off, meaning the firmware is updated completely.

更新完成後，A13 GST遙控器會自動關機，代表主程式已更新完畢。

ALIGN

3 FUNCTION SETUP
功能設定

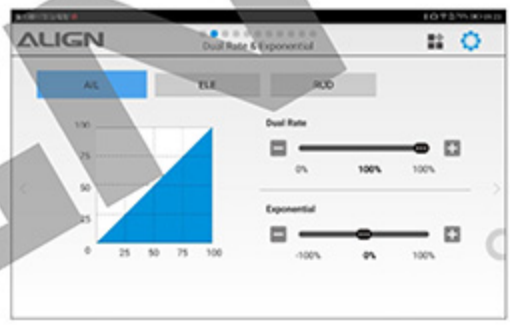
Enter system menu, select function setup or slide the screen to enter function setup page. Some functions have different settings based on different model selection.

進入系統選單，選擇想要的功能，點按或滑動畫面進入功能設定頁面。部分功能會因選擇模型不同而有不同的功能設定。



1 DUAL RATE & EXPONENTIAL
雙重比率&動作曲線

DUAL RATE : Commonly abbreviated as "D/R", dual provides the ability to change the travel distance of a channel (servo) thus affecting the overall travel of the servo(s). Decreasing the percentage value for a given servo will reduce the travel amount of the channel respectively. For example, if the rudder of the helicopter travels 1/2" at the high rate and maximum stick input, and a value of 50% is input for the low rate when the switch is moved to the low rate position, the servo moves precisely half as far per stick position.example, if the rudder control input is all the way deflected, the maximum travel is now 1/4".





EXPONENTIAL: Commonly abbreviated as "EXP" is used to adjust the sensitivity of center stick the neutral point. Adjusting "EXP" does not affect the travel range. For example: When EXP set at positive, the neutral point will be less sensitive and slow. When EXP set at negative, the neutral point will be sensitive and very agilty.



雙重比率：D / R是雙重比率的縮寫簡稱，這功能可改變伺服器的行程量，然後影響伺服器的輸出。因此，減少伺服器雙重比率的百分比將會減少頻道輸出的行程量。例如：如果開關在高比率位置且撥桿為最大輸入時，對應的尾舵的輸出行程為1/2"(英吋)，當開關切到低比率且數值設定為50%，則尾舵的行程就會變成原來的一半。在這個例子中，開關切換到低比率時，尾舵的最大行程會變成1/4"(英吋)。

動作曲線：簡稱EXP，主要影響動作中立點位置的輸出反應，EXP不會影響到伺服器的行程量大小，當EXP數值設定為"+"時，該動作中立點附近反應會變緩慢；反之，設定為"-"時該動作中立點附近反應會變快變靈敏。

ALIGN

2 SERVO TRAVEL ADJUSTMENT
伺服器行程量調整

End Points for all channels (servos) can be adjusted to improve flight control based on the model.
The end point can be adjusted from 0%~125%. By setting the maximum end point, this can protect the servo from over travel and damage.

調整各種伺服器及頻道範圍的大小量，以便伺服器輸出及頻道數據輸出符合結構設計及性能的要求，達到最佳的控制效果。

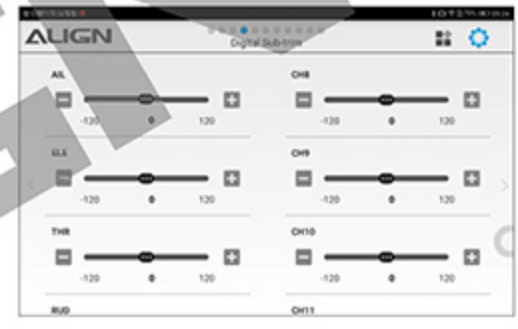
行程量限制點可以從0%~125%之間調整。設定此限制點後，即使混控等操作使得伺服器行程量增加，伺服器的動作也不會超越限制點，進而保護伺服器的鎖定及連接作用。



3 DIGITAL SUB-TRIM
數位微調

Sub-trim can be used on CH1~CH10 to change the channel/servo neutral position. For example; if the RC model drifts slightly, the sub-trim can correct this flight issue.

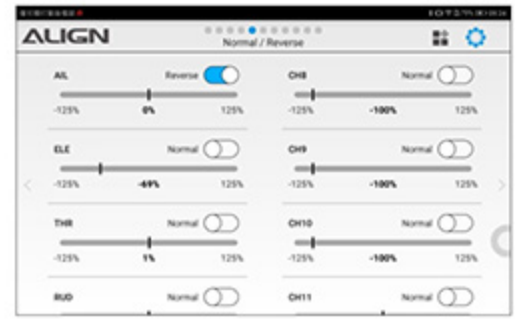
微調功能用來調整CH1~CH10頻道/伺服器的中立點，當伺服器因結構固定的角度差無法調整時，可透過微調調整每個頻道對應伺服器的中立點。例如：如果遙控模型的中立點位置輕微偏移，可透過微調能來修正該問題。



4 NORMAL / RESERVE
通道正反向

Reverse provides the ability to change each channel (servo) direction. It may be necessary to change the servo direction based upon the specific model requirements.

此功能針對不同類型的伺服器與其頻道輸出方向或是各種機種的安裝方式不同，調整伺服器及頻道的輸出方向，可使模型上所有伺服器的方向透過此設定改變。



CAUTION 注 意

Always double check all servos to confirm transmitter stick movement is working correctly based on the model.

操控時，請務必再次確認所有伺服器的動作方向與操控方向一致。

ALIGN

5 THROTTLE CURVE
油門曲線

Allows the ability to adjust the Throttle Curve based on the model and flight requiements.

The throttle curve has five adjustable points; L=0%, 1=25%, 2=50%, 3=75%, H=100% throttle stick movement. Each point can be adjusted from 0%~100% for throttle range based on motor, pitch curve, and flight requirements.

此功能用來調節油門的操作曲線，使搖桿動作與油門的相互協調，達到理想之油門輸出反應。

油門曲線提供5個主要位置點可調整，分別對應油門搖桿的最低、1/4、1/2、3/4、最高位置，每點可調整油門範圍為0%~100%，依照馬達引擎特性與螺距曲線(直昇機)搭配，匹配最佳油門曲線位置。





6 PITCH CURVE
螺距曲線

Suitable Models： Helicopter
適用模型類型：直昇機

Pitch Curve is used for a Helicopter model. Offer five adjustable points; L=0%, 1=25%, 2=50%, 3=75%, H=100% stick travel. Each point can be adjusted from 0%~100% based on throttle range and desired pitch requirements of the helicopter model.

螺距曲線提供5個主要位置點可調整，分別對應油門搖桿的最低、1/4、1/2、3/4、最高位置，每點可調整油門範圍為0%~100%，依照您所想要的螺距輸出並匹配油門曲線來設定。





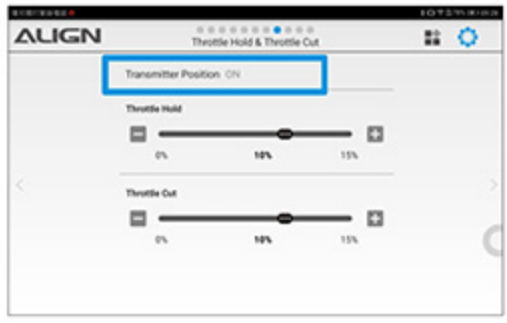
ALIGN

7 THROTTLE HOLD & THROTTLE CUT
油門鎖定&油門熄火

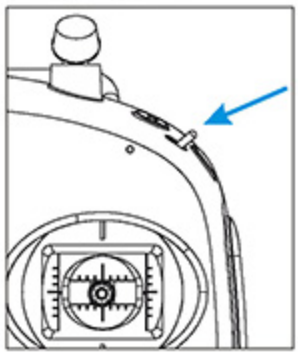
Suitable Models： Helicopter
適用模型類型：直昇機

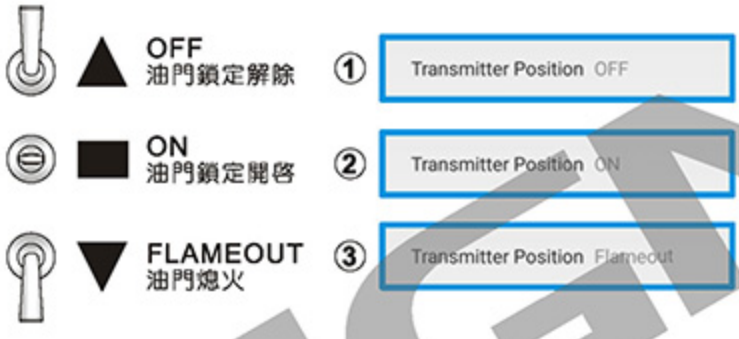
Allow setting throttle hold and throttle cut (engine cut). Throttle hold is a security setup, limiting motor or engine operation before flight or cut motor power immediately, or activate when landing.

此功能可以設定油門鎖定、油門熄火(引擎熄火)兩種功能。
油門鎖定是個安全設定，可以在飛行前限制馬達或引擎的運作與突然切斷馬達動力，或是執行直昇機熄火降落時使用。



Transmitter Position
遙控器目前位置





CAUTION 注 意

Always double check all servos to confirm transmitter stick movement is working correctly based on the model.

操控時，請務必再次確認所有伺服器的動作方向與操控方向一致。

8 GYRO GAIN
陀螺儀感度

Suitable Models： Helicopter
適用模型類型：直昇機

Suitable for a Helicopter model. Allow setting tail gyro gain of three flight mode.

可設定三種飛行模式的尾舵陀螺儀感度值



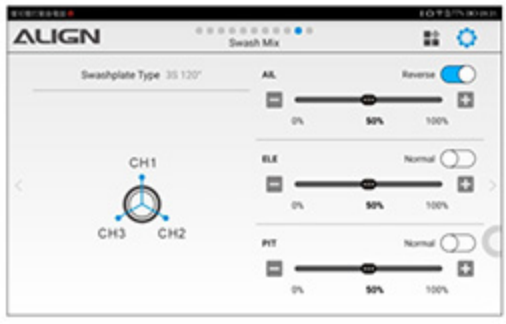
ALIGN

9 SWASH MIX
十字盤動作行程量

Suitable Models：Helicopter (3S 120°)
適用模型類型：直昇機 (3S 120°)

Allow correct the swashplate in the aileron and elevator or direction corresponding to each operation of each condition. For example; change "reverse" switch to reverse the swashplate direction.

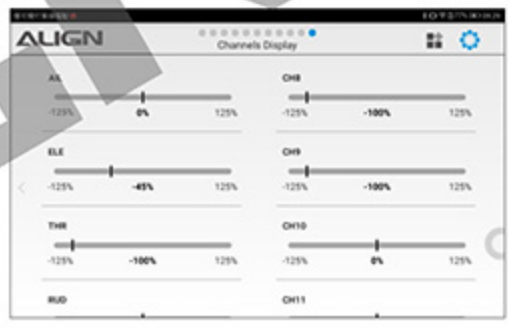
此功能是設置十字盤副翼、升降和螺距的行程動作量。若需要使十字盤動作反向，必須調整該動作右上方正反向開關



10 CHANNELS DISPLAY
各頻道顯示資訊

Display all channels instant operation data, also can take advantage of this function to confirm channel outputs are correct.

顯示所有頻道即時的操控資料，也可利用該功能來確認遙控個動作輸出是否正確。



GROUND MONITORING AND MANAGEMENT SYSTEM
地面監控管理系統

To use Ground Monitoring and Management System, A13 GST transmitter has to be with AP1 Multicopter Flight Controller or AP2 Flight Control System, VTD1 1080 HD Digital Video Transmitter Set, G2P Gimbal.

The Ground System APP simplifies route mapping and automated flight. Support available for iOS or Android devices such as phones and tablets. Parameters and flight settings can be adjusted anytime or anywhere with the intuitive on-screen interface. A complex mission can quickly be programmed in way points setup by simply press the screen to achieve full automation of mission.

A13 GST 遙控器須搭配AP1多軸飛行控制器或AP2飛行控制器、VTD1 1080高清數位圖傳、G2P雲台DV組即可擁有地面監控管理之功能。

透過APP實現自主航線規劃及飛行。支援iOS、Android系統手機/平板，可以隨時隨地進行設定與參數調整，擁有直觀簡易的操作畫面，只需輕點螢幕，就能輕鬆規劃複雜航線任務。

ALIGN

1 VTD1 1080 HD DIGITAL VIDEO TRANSMITTER SET ASSEMBLY ILLUSTRATION
VTD1 1080高清數位圖傳組組裝示意圖

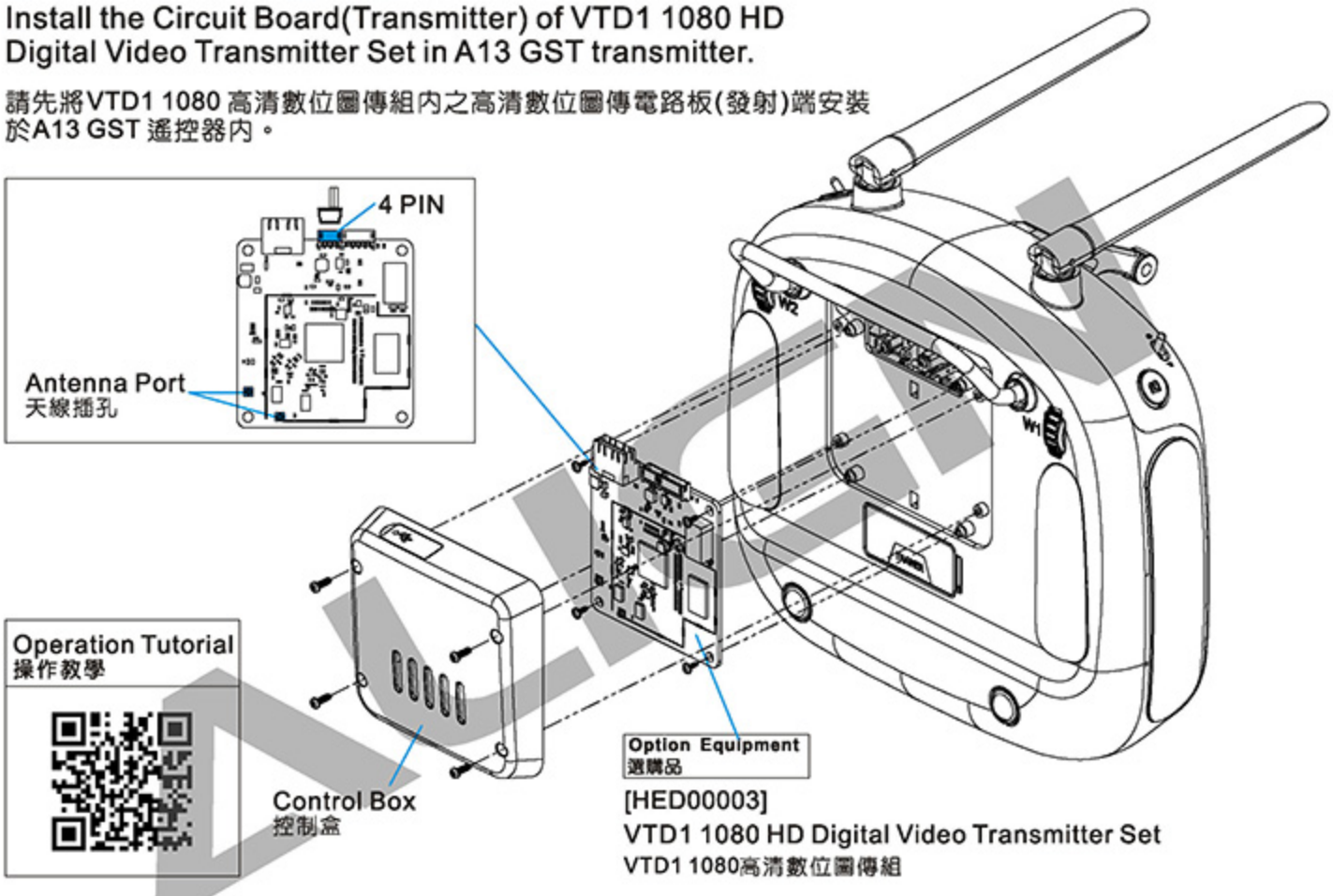
CAUTION 注 意

VTD1 1080 HD Digital Video Transmitter Set is bound at the factory. If either the receiver or ground launcher is changed, the aircraft requires rebinding.

VTD1 1080高清數位圖傳組出廠前已完成對頻。若必須更換空中接收端或地面發射端其中一端設備時，才需重新進行對頻。

Install the Circuit Board(Transmitter) of VTD1 1080 HD Digital Video Transmitter Set in A13 GST transmitter.

請先將VTD1 1080 高清數位圖傳組內之高清數位圖傳電路板(發射)端安裝於A13 GST 遙控器內。



2 SOFTWARE DOWNLOAD AND INSTALLATION
APP 下載安裝

Please rest assured that the Flight Controller is the latest version before leaving factory. You can connect to Align official website to update the latest version and any news. Please scan QR Code to download related software from Align website or search it in iOS/Android App store.

Please download the Flight Controller App of your mode and follow steps to finish installation and settings.

飛行控制器，在出廠前已是最新版本，請安心使用。您也可以連結至亞拓官網查詢，隨時更新亞拓發佈的最新版本及各項最新訊息。請掃描QR Code連結亞拓網站下載相關軟體。或是在iOS/Android App store 搜尋。

請依您所使用的模型機型下載飛行控制器APP，依循下列步驟安裝調校完成。

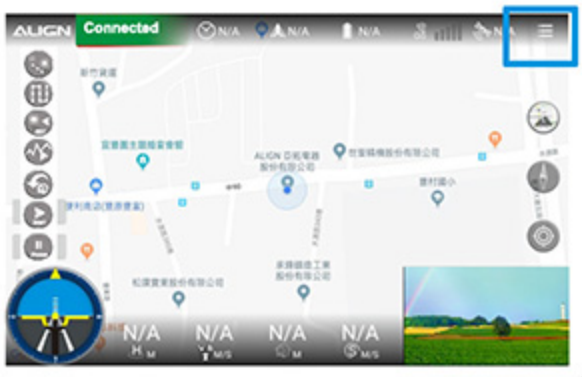
1 MULTI-DRONE (M470L/M480/M690) DOWNLOAD AND INSTALLATION
多軸無人機 (M470L/M480/M690) 安裝教學

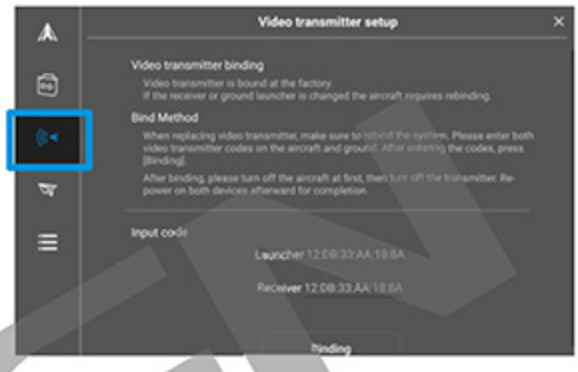
www.align.com.tw/download-en/AP1/

Compatible with ANDROID APP ON Google play | Download on the App Store

Open AP1 APP, click "Universal Settings" and enter "Video Transmitter Setup" to proceed with binding.

請開啓AP1 APP，點選〔通用設置〕開啓〔圖傳設置〕，進行圖傳對頻。





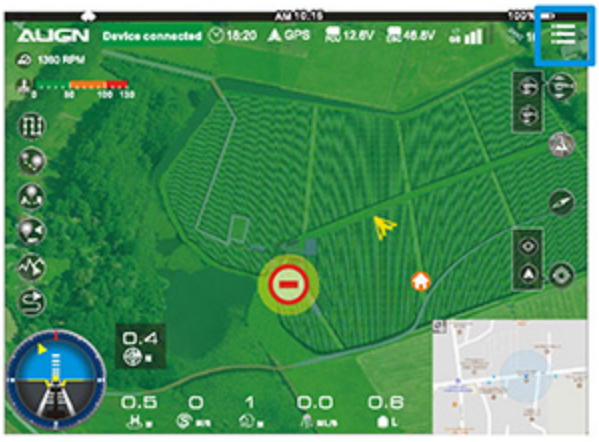
2 AGRICULTURAL HELICOPTER (ALIGN E1/ G1) DOWNLOAD AND INSTALLATION
植保直昇機 (ALIGN E1/ G1) 安裝教學

www.align.com.tw/download-en/AP2/

Compatible with ANDROID APP ON Google play | Download on the App Store

Open AP2 APP, click "Universal Settings" and enter "Video Transmitter Setup" to proceed with binding.

請開啓AP2 APP，點選〔通用設置〕開啓〔圖傳設置〕，進行圖傳對頻。





ALIGN

3 VTD1 1080 HD DIGITAL VIDEO TRANSMITTER SET BINDING INTRODUCTION
VTD1 1080 高清數位圖傳組對頻方式

BIND METHOD :

A.) When replacing video transmitter, make sure to rebind the system. Please enter both video transmitter codes on the aircraft (receiver code) and ground (launcher code). After entering the codes, press [Binding].

B.) After binding, please turn off the aircraft at first, then turn off the transmitter. Re-power on both devices afterward for completion.

對頻方式：

A)當更換任一端圖傳設備時，以空中接收端所使用的圖傳組編碼為主，對應同一組地面發射端編碼，輸入該組編碼後，按〔對頻〕。

B)對頻完成後，先關閉飛行機電源再關閉遙控器電源，然後重新開機即完成。

VIDEO TRANSMITTER CODES DIAGRAM　圖傳編碼示意圖

Each video transmitter code is unique. Please refer to below diagram and enter the code.

每套圖傳都有獨立且對應的編碼序號，請參考圖選位置，輸入您圖傳的專屬編碼。



Launcher Code
發射端編碼

80201-0018-A
12:DB:01:AA:01:C9
Align-018

Receiver Code
接收端編碼

80201-0018-B
12:DB:01:BB:01:C9

A13 GST TRANSMITTER FIRMWARE UPDATE
A13 GST遙控器主程式下載更新

ALIGN

A13 GST Transmitter default program is the latest version, please feel at ease to use. You can also link to ALIGN website to get the latest version and the latest news.

A13 GST遙控器，在出廠前已是最新版本，請安心使用。您也可以連結至亞拓網站查詢，隨時更新亞拓發佈的最新版本及各項最新訊息。

1.PC firmware update: Download A13 GST transmitter update firmware refering to below link:http://www.align.com.tw/A13

電腦主程式更新：下載A13 GST遙控器主程式更新安裝檔，請至下列網址
http://www.align.com.tw/A13

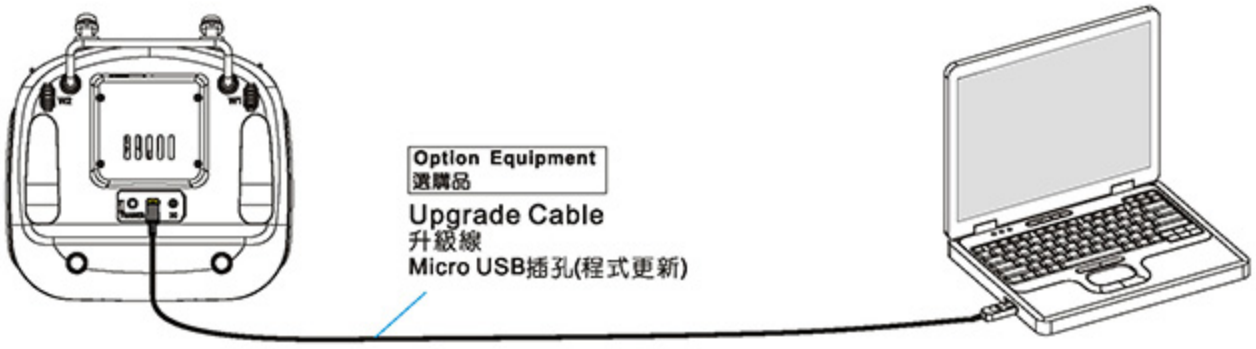
Scan QR CODE and read the detail information for A13 GST Transmitter Firmware Update and Instruction.

A13 GST遙控器主程式下載更新
請掃描QR CODE會有完整的教學步驟，請依教學步驟完成設定安裝。

ALIGN

2.Turn on A13 GST transmitter, and unzip the upgraded software, and then connect upgrade cable to the computer.

請先開啓A13 GST遙控器，將升級軟體壓縮檔，進行解壓縮。解壓縮完成後，請將A13 GST遙控器升級線(選購品)連接至電腦。



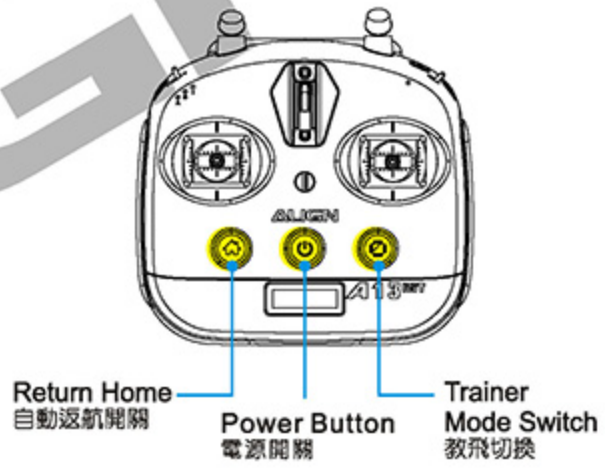
CAUTION 注 意

Please double check upgrade cable is inserted correctly, or can't be upgraded completely.

請再次檢查升級線是否安插正確，沒有確實安插則無法下載更新。

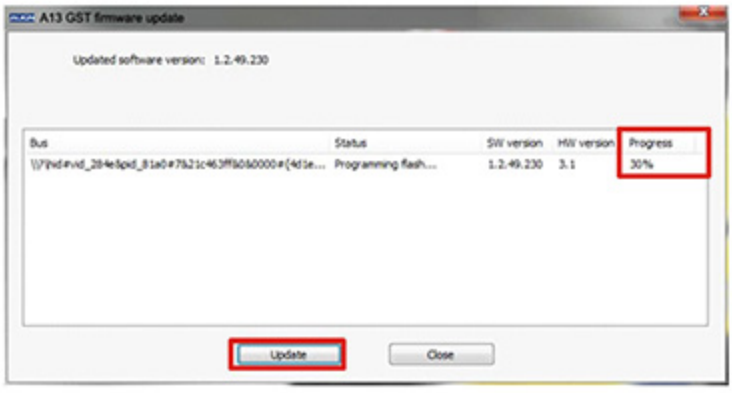
3.After confirm if upgraded cable connected correctly, press auto return button, power button and trainer mode switch at the same time. Please refer to the yellow button in the right image. Then, the sound will be heard when turning on A13 GST transmitter.

確認連接升級線正確無誤，同時按下(自動返航開關、電源開關、教飛切換)請參閱右圖標示黃色按鈕部分，開機，此時A13 GST遙控器會有開機聲響。



4.Click the file which has been unzipped, ALIGN A13 GST firmware update screen will be popped up, please press "Update" to install it. Installation progress can be viewed under the "Progress" option.

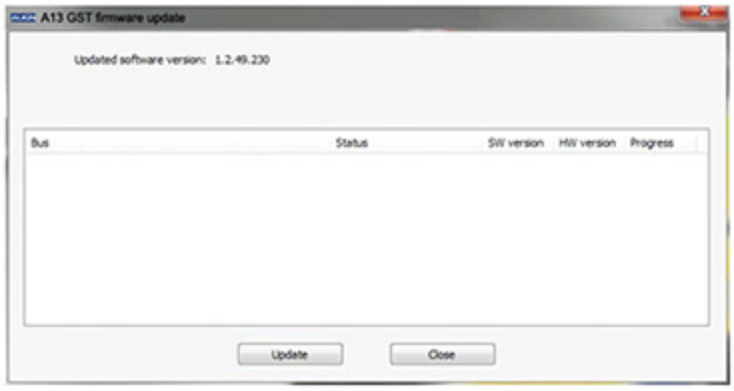
點按已解壓縮完成的檔案，系統會自動跳出ALIGN A13 GST firmware update的安裝畫面，請點按" Update 進行安裝。安裝進度可於" Progress "選項下方檢視。



ALIGN

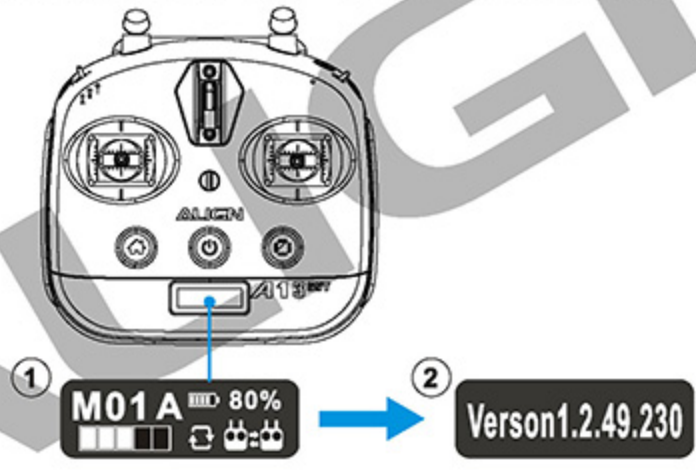
5.After completion, window information will disappear. A13 GST transmitter will turn off automatically and make a sound, which means the installation is complete.

安裝完成後，視窗的資訊會消失。此時A13 GST遙控器會自動關機並發出關機聲響，則代表已安裝完成。



6.Remove upgraded cable, and extended press power switch of A13 GST transmitter, and short press power switch again to show current version. It will be shown according to the version you upgraded, the image is only for reference.

接著將升級線移除，長按A13 GST遙控器電源開關開機，再短按一次A13 GST遙控器電源開關則會出現目前安裝的版本。安裝版本會依您實際更新之版本呈現，圖片僅供示範參考用



PRE-FLIGHT CHECKLIST AND WARNINGS
飛行前檢查與注意

ALIGN

Prior to every flight, carefully check all parts such as blades, screws, frame, arms, etc; ensure they are firmly secured and show no unusual wears, or unforeseen danger may happen.

Make sure to double check all APP parameter setting before flying. Incorrect settings will result in unexpected and unpredictable safety danger.

每趟飛行前須仔細檢查機身各部位之零/配件/電子設備性能是否正常，及無損耗老化現象，並確實將螺絲鎖緊才能升空飛行。做好定期檢修，避免零件或電子產品異常所造成不可預期的意外。

飛行前務必確認APP內所設定的參數是否正確，不正確的設定將會影響飛行狀態的安全，嚴重將導致不可預期的危險。

Make sure to do all safety check carefully and seriously.
Wish you a joyful flight experience.

飛行前做好各項安全檢查才能有更美好的飛行狀態，
祝福您有一個愉快的飛行體驗。

PRODUCT SPECIFICATION
產品規格

ALIGN

A13 GST TRANSMITTER SPECIFICATION　A13 GST遙控器規格

Contents 內容	Specifications 規格
Output Channel 通道數量	14 channels
Model Type 適用機型	Helicopter/ Multicopter 直昇機 / 多軸機
RF Range 頻率範圍	2.408GHz~2.475GHz
RF 發射功率	<20dBm(For EU)
System Mode 系統模式	2.4G system: AFHDS
Power 電源	Built-in 2S 7.4V 5800mAh Li-Polymer Battery 內建 2S 7.4V 5800mAh充電式鋰電池
Display 顯示方式	128x64mm
Transmit Distance 發射距離	In the air without interference ＜1000 meters 空曠無干擾空中距離：＜ 1000 米
Weight 重量	650g
Dimension: (L x W x H) 尺寸	167x180x103mm
Certificate 認證	CE,FCC
Bluetooth Work Frequency Band 藍牙工作頻段	2402~2480MHz
Bluetooth Max. Power 藍牙最大功率	<8dBm (Declaration for EU Compliance)

A6B RECEIVER SPECIFICATION　A6B接收器規格

Contents 內容	Specifications 規格
Output Channel 通道數量	14 channels
RX Sensitivity 接收器靈敏度	-105dbm
RF Range 頻率範圍	2.408GHz~2.475GHz
System Mode 系統模式	2.4G system: AFHDS
Power 電源標準	4.0~8.4V DC
Receive Distance 接收距離	In the air without interference ＜1000 meters 空曠無干擾空中距離：＜ 1000 米
Weight 重量	14g
Dimension: (L x W x H) 尺寸	46.7x25.5x13.6mm
Certificate 認證	CE,FCC

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Thank you for purchasing and supporting ALIGN products.

The Align Team is dedicated to you by innovating and developing new RC Helicopters, Multicopters, and FPV Racing Quads. We strive to provide a more diversified experience for our customers. Visit our website at www.align.com.tw for latest news, information, and updates about our extensive line of products for the RC enthusiasts.

Good Flying!



再次感謝您對亞拓系列商品的喜愛與支持，您的肯定是對我們最大的認同。

亞拓團隊秉持創新研發的精神，開發遙控直昇機／多軸飛行機／穿越機系列商品，提供給您體驗更多樣化的飛行樂趣。您可以透過下列連結，隨時瞭解亞拓的最新動態，以及各項訊息分享。

祝福您有一個愉快的飛行體驗。

ALIGN Flight Safety 亞拓飛行安全宣導
http://www.align.com.tw/flysafe-en/



ALIGN T-REX Helicopter 亞拓遙控直昇機
http://www.align.com.tw/helicopter-en/

ALIGN Multicopter 亞拓多軸飛行機
http://www.align.com.tw/multicopter-en/



ALIGN FPV Racing Quad 亞拓穿越機
http://www.align.com.tw/multicopter-en/mr25/

ALIGN Website 亞拓官網
http://www.align.com.tw

ALIGN Shopping Cart 亞拓購物車
http://shop.align.com.tw/index.php?language=en

ALIGN Quick Finder 亞拓零件快速購
http://shop.align.com.tw/partfinder.php?language=en

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https://www.facebook.com/Align-Corporation-194493419543/?ref=mf

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https://instagram.com/aligncorporation/

ALIGN YouTube
https://www.youtube.com/channel/UCaPj_K5DNo7HSmP1eytUvMQ

ALIGN Youku
http://i.youku.com/u/UMTQ0NjEwNjczNg==

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